What a Solution Remembers About Its Equation: Groups Are Factors, Terms Interact, and Where Exactness Ends

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Abstract. Given one exact solution $u(t,x)$ of an unknown linear equation with constant coefficients, what does u remember about the equation? We prove that the right unit is not the additive term but the irreducible *factor* of the symbol. (i) u determines its operator P exactly when no partial quotient $(P/P_i)(\partial)u$ is an exponential-polynomial; no finite amount of data can certify this, and a black-box test is impossible, yet it becomes decidable under a modest extra hypothesis. (ii) On a simply connected region u splits into one component per factor of the symbol, uniquely up to an exponential-polynomial overlap of known finite dimension; a split per additive term is in general impossible; the split is stable exactly when the sum of the factors' solution spaces is closed, which for two factors means a positive Friedrichs angle. (iii) Inside a factor, for evolution equations $u_t = \sum_n A_n u$ with commuting terms, the part of the solution due to the interplay of the terms is $\prod_n(\mathrm{Id}+\Delta_n) - \mathrm{Id} - \sum_n \Delta_n$ applied to the datum; it costs $O(N)$ per value but $\Theta(2^N)$ as a table, and it is not continuous in Sobolev norms. (iv) Nonlinear equations such as Burgers admit analogous statements, with sharp asymptotics across a shock. Every statement carries a verification status. Each concept is illustrated by a worked example, and a primer for readers new to ideals and companion matrices is included.

Key words. differential operators, annihilator ideal, factorization, exponential-polynomials, interaction term, Friedrichs angle, Burgers equation

MSC codes. 35E20, 35A25, 13P25, 35Q53

1 Introduction

Consider a quantity that is carried by a flow and diffuses at the same time. A natural model is

$$(\partial_t + c\,\partial_x)(\partial_t - \nu\,\partial_x^2)\,u = 0. \tag{1}$$

Its solutions are sums of a travelling profile and a diffusing one. Suppose you are handed one such solution $u(t,x)$, known exactly on a region of the plane, and are not told the equation. Three questions arise, and this paper answers all three.

- **Q1. Identification.** Can the operator be reconstructed from u, and when is it unique?
- **Q2. Decomposition.** If the operator is built from pieces, can u be split into the parts each piece generates? For (1): is there a travelling part and a diffusing part, and what are they?
- **Q3. Interaction.** If the pieces are the *terms* of the operator (advection, diffusion, ...), what part of u is due to no single term but to their interplay, and can it be computed exactly and cheaply?

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(a) oscillation (b) damping

Figure 1: The idea of the paper on the equation $(\partial_t + c\,\partial_x)(\partial_t - \nu\,\partial_x^2)u = 0$. Every Fourier mode $e^{\lambda t + i\kappa x}$ of a solution lies on the zero set of one factor of the symbol $(\tau + c\xi)(\tau - \nu\xi^2)$: circles on the line of the transport factor, squares on the parabola of the diffusion factor (schematic). Identifying the equation means finding these two groups.

The setting is deliberately analytic: no noise, no statistical model. The aim is an *exact* statement and a precise account of what is impossible.

The idea in one paragraph In Fourier variables a constant-coefficient operator is a polynomial in (τ, ξ): that of (1) is $(\tau + c\,\xi)(\tau - \nu\,\xi^2)$. Every Fourier mode $e^{\lambda t + i\kappa x}$ of a solution lies on the zero set of one factor: the modes of the first factor on the line $\lambda = -ic\kappa$, those of the second on the parabola $\lambda = -\nu\kappa^2$ (Fig. 1). The modes therefore fall into groups, and the groups are the *factors*, not the terms $(\partial_t^2,\ c\nu\,\partial_x^3,\ \dots)$ of the expanded operator. Everything that follows makes this precise.

How to read the status labels Each statement carries one of: [proved] (complete proof, independently reviewed), [proved mod. X] (complete except for a cited published theorem X whose source was checked), [conditional] (depends on an unverified lemma), [open], or [false] (a counterexample exists). The reviewers were independent AI agents, not human referees; fifteen core algebraic theorems were additionally machine-checked in Lean 4 (branch directory `proofs/lean`). Grey tags such as (ledger C3-T1) refer to entries of the claim ledger kept with the proofs.

Results at a glance Table 1 lists the main statements; Fig. 2 shows how they depend on each other. Section 2 builds the algebra, Section 3 proves the grouping principle and its stability, Section 4 treats the interaction term, Section 5 asks when a solution determines its equation, Section 6 moves to nonlinear equations, and Section 14 collects the limits. Proofs are in Appendix A; Appendix C is a primer.

2 Derivatives become polynomials

In brief. Before asking what a solution knows, we turn operators into polynomials and solutions into objects that carry an ideal of equations; the one-variable case shows every idea in

Table 1: Main results in the order in which they are proved. "Proved" means a complete proof that survived an independent review.

Topic	Result	Status
Identification	$\mathrm{Ann}(u) = (P) \iff (P/P_i)(\partial)u \notin \mathrm{EP}$ for every irreducible $P_i \mid P$ (Theorem 5.1)	[proved]
Data limits	no finite certificate; black box impossible; decidable with a bound or a holonomic model (Proposition 5.5)	[proved]
Decomposition	$\ker P = \sum_j \ker Q_j$ on simply connected Ω, overlap of dimension $\delta(P)$ (Theorem 3.1)	[proved]
Terms vs. factors	a split per additive term is false in general (Proposition 3.3)	[proved]
Stability	the split is stable iff $\gamma_N > 0$; for two factors, positive Friedrichs angle (Theorem 3.4)	[proved]
Interaction term	$E_{\mathrm{int}} = \prod(\mathrm{Id} + \Delta_n) - \mathrm{Id} - \sum \Delta_n$; $O(N)$ value, $\Theta(2^N)$ table (Theorem 4.1)	[proved]
Instability	$u_0 \mapsto E_{\mathrm{int}}$ is not continuous in H^m (Proposition 4.2)	[proved]
Nonlinear	Burgers: closed form before the shock, sharp behaviour across it (Section 6)	[proved]/ [proved mod. Kružkov]

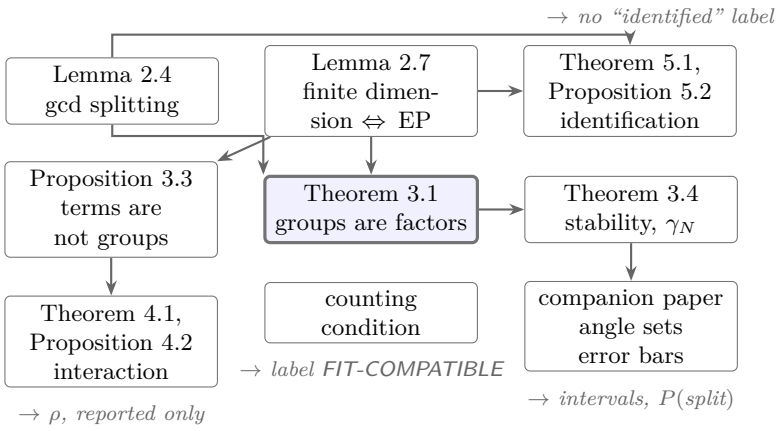


Figure 2: How the results of Section 2 depend on each other (arrows) and where the algorithm uses them (italic). The shaded result is the grouping principle; everything else either prepares it or measures how well it can be carried out on finite data.

miniature.

2.1 Derivatives become polynomials

Throughout, $\Omega \subset \mathbb{R}^2$ is a domain in the variables (t, x) and every operator has constant coefficients,

$$P(\partial_t, \partial_x) = \sum_{j,k} c_{jk}\, \partial_t^j \partial_x^k, \qquad c_{jk} \in \mathbb{C}.$$

The exponentials $\psi_{\tau,\xi}(t, x) = e^{\tau t + \xi x}$, $(\tau, \xi) \in \mathbb{C}^2$, are common eigenfunctions of all such operators: $\partial_t \psi_{\tau,\xi} = \tau \psi_{\tau,\xi}$, $\partial_x \psi_{\tau,\xi} = \xi \psi_{\tau,\xi}$ and therefore

$$P(\partial)\, \psi_{\tau,\xi} = P(\tau, \xi)\, \psi_{\tau,\xi}. \tag{2}$$

Differentiation is thus replaced by multiplication, and operators are replaced by polynomials.

Definition 2.1 (Operator ring, symbol, characteristic variety)**.** The constant-coefficient operators form the commutative ring $\mathcal{R} = \mathbb{C}[\partial_t, \partial_x] \cong \mathbb{C}[\tau, \xi]$. The polynomial $P(\tau, \xi)$ is the *symbol* of $P(\partial)$, and its zero set $V(P) = \{(\tau, \xi) \in \mathbb{C}^2 : P(\tau, \xi) = 0\}$ is the *characteristic variety*.

By (2), $\psi_{\tau,\xi}$ solves $P(\partial)u = 0$ exactly when $(\tau, \xi) \in V(P)$. A Fourier mode in space is the case $\xi = i\kappa$ with wavenumber $\kappa \in \mathbb{R}$; then $P(\tau, i\kappa) = 0$ is the *dispersion relation*, and the roots $\tau = \lambda(\kappa)$ are the temporal rates of the mode $e^{\lambda(\kappa)t + i\kappa x}$.

Convention We keep the algebraic variable ξ in statements about the ring and write $\xi = i\kappa$ only when we evaluate on Fourier modes. With this convention a symbol has real coefficients exactly when the equation is real, and no factor i ever appears inside $\mathcal{R}$.

Example 2.2 (Transport plus diffusion: the running example)*.* The operator $P = (\partial_t + c\,\partial_x)(\partial_t - \nu\,\partial_x^2)$, with speed $c > 0$ and diffusivity $\nu > 0$, has symbol

$$P(\tau, \xi) = Q_1 Q_2, \qquad Q_1 = \tau + c\,\xi, \qquad Q_2 = \tau - \nu\,\xi^2.$$

On Fourier modes the two factors give two dispersion branches, $\lambda_1(\kappa) = -ic\kappa$ (a wave that travels without decay) and $\lambda_2(\kappa) = -\nu\kappa^2$ (a standing mode that decays). A solution is any superposition of modes on the two branches. The equation is one object; its *two factors* are two groups of modes. Recovering those groups from data is the subject of this paper. We use $c = 1$, $\nu = 0.3$ in all numerical illustrations.

2.2 What a solution knows: the annihilator

A single function satisfies many equations: e^x is annihilated by $\partial_x - 1$, by $\partial_x^2 - 1$ and by every multiple of $\partial_x - 1$.

Definition 2.3 (Annihilator)**.** For a distribution u on Ω, $\mathrm{Ann}(u) = \{Q \in \mathcal{R} :\ Q(\partial)u = 0\}$.

$\mathrm{Ann}(u)$ is an ideal: it is closed under sums and under multiplication by any operator, because $(SQ)(\partial)u = S(\partial)(Q(\partial)u) = 0$. It collects every constant-coefficient equation that u satisfies. We use three classical facts about $\mathbb{C}[\tau, \xi]$ [1]: unique factorization into irreducible polynomials; Bézout's theorem (two polynomials without a common factor have finitely many common zeros); and Hilbert's Nullstellensatz (if q vanishes on the zero set $V(I)$ of an ideal I, then $q^k \in I$ for some k). All three require the complex field, which is why symbols are always considered over $\mathbb{C}$.

Lemma 2.4 (Splitting off the greatest common divisor)**.** *Every nonzero ideal $I \subset \mathcal{R}$ factors as $I = g \cdot J$, where $g = \gcd(I)$ and J is an ideal with finite zero set $V(J)$.*

The proof (Appendix A) is elementary: divide by g, then use Bézout on two elements of J without a common factor. Applied to $I = \operatorname{Ann}(u)$ the lemma gives the decomposition

$$\operatorname{Ann}(u) = G_u\, J_u, \qquad G_u = \gcd \operatorname{Ann}(u), \qquad \Sigma_{\mathrm{p}}(u) = V(J_u) \text{ finite.} \tag{3}$$

The polynomial G_u is the part of the equation that u fills out along whole curves of $\mathbb{C}^2$; the finite set $\Sigma_{\mathrm{p}}(u)$, the *point spectrum* of u, lists the isolated exponentials that u happens to contain. We call G_u the *intrinsic operator* of u: it depends on u alone, whereas an annihilator P supplied from outside may contain spurious factors.

Reading the two pieces The two objects in (3) answer different questions. The polynomial G_u is a differential equation: its zero set $V(G_u)$ is a union of curves in $\mathbb{C}^2$, and each curve is a continuous family of exponential modes $e^{\tau t+\xi x}$, all solving $G_u(\partial)v = 0$. The point spectrum $\Sigma_{\mathrm{p}}(u)$ is a list: the finitely many individual exponential modes that are present in this particular function u and are not accounted for by G_u. In short, G_u identifies an equation that describes a whole family of modes; $\Sigma_{\mathrm{p}}(u)$ identifies the isolated modes that u happens to contain. Example 2.5 shows the two extreme cases.

Example 2.5 (Two extremes). For $u = e^x$, $\operatorname{Ann}(u) = (\tau,\ \xi - 1)$ is the ideal of the single point $(0, 1)$; hence $G_u = 1$ and $\Sigma_{\mathrm{p}}(u) = \{(0, 1)\}$: the function knows one exponential and no equation. For $u = f(x - ct)$ with a profile f that is not a finite sum of exponentials times polynomials, $\operatorname{Ann}(u) = (\tau + c\xi)$; hence $G_u = \tau + c\xi$ and $\Sigma_{\mathrm{p}}(u) = \emptyset$: the function knows its equation completely (Theorem 5.1).

2.3 Exponential-polynomials: the solutions that remember too little

Definition 2.6 (Exponential-polynomials)**.** EP is the set of finite sums $\sum_j p_j(t, x)\, e^{\tau_j t+\xi_j x}$ with polynomials p_j and exponents $(\tau_j, \xi_j) \in \mathbb{C}^2$. $\mathrm{EP}(\Omega)$ denotes their restrictions to Ω.

For a function w let $\mathcal{R}w = \operatorname{span}\{\partial_t^j \partial_x^k w :\ j, k \geq 0\}$ be the space of all its derivatives. The map $Q \mapsto Q(\partial)w$ is onto $\mathcal{R}w$ with kernel $\operatorname{Ann}(w)$, so $\mathcal{R}w \cong \mathcal{R}/\operatorname{Ann}(w)$ as vector spaces.

Lemma 2.7 (Finite-dimension criterion)**.** *Let Ω be connected. Then $w \in \mathrm{EP}(\Omega)$ if and only if $\dim_{\mathbb{C}} \mathcal{R}w < \infty$. In that case $V(\operatorname{Ann} w)$ is the finite set of exponents $\{(\tau_j, \xi_j)\}$ that occur in w.*

Proof idea. We show that w is an exponential-polynomial if and only if its derivatives span a finite-dimensional space. ($\Rightarrow$) For one term $w = p(t, x)\, e^{at+bx}$,

$$\partial_t w = (p_t + a\, p)\, e^{at+bx}, \qquad \partial_x w = (p_x + b\, p)\, e^{at+bx}.$$

Differentiation never raises the degree of p, so every derivative of w lies in the finite-dimensional space of polynomials of degree at most $\deg p$ times e^{at+bx}; a finite sum of such terms stays in a finite sum of such spaces. ($\Leftarrow$) If $\mathcal{R}w$ is finite-dimensional, ∂_t and ∂_x map it to itself and act on it as two commuting matrices A and B. Their minimal polynomials m_A and m_B give two constant-coefficient differential equations,

$$m_A(\partial_t)\, w = 0, \qquad m_B(\partial_x)\, w = 0,$$

one in each variable. On every rectangle inside Ω their common solutions are exponential-polynomials, and connectedness extends this representation to all of Ω. The joint eigenvalues (a, b) of A and B are the exponents that occur in w. Details are in Appendix A. □

In brief. Deciding whether $w \in \mathrm{EP}$ is a finite-dimensional linear-algebra question: compute the rank of the derivatives of w. The joint eigenvalues of the two matrices are the exponents, that is, the points of $\mathbb{C}^2$ at which w lives. This is the computational core of everything that follows: the matrix pencil of the companion algorithm paper is exactly this eigenvalue computation, done for ∂_t one wavenumber at a time.

Example 2.8 (Rich and poor solutions of the heat equation). A single heat mode $e^{-\nu\kappa^2 t + i\kappa x}$ and any finite sum of such modes belong to EP. The heat kernel $t^{-1/2}e^{-x^2/(4\nu t)}$ does not: its x-derivatives are polynomials of increasing degree times the same Gaussian, so they span an infinite-dimensional space.

Starting from: one variable x, an ODE $p(D)u = 0$ with constant coefficients, $D = d/dx$, p known. *Why:* in one variable every object of the later sections (minimal annihilator, spectral projectors, companion matrix) appears with explicit formulas and no geometry of curves. *Role:* prototype for Section 5.1 (identification) and Section 3 (the Lagrange projectors reappear as Eq. (11)).

This section introduces, in the simplest setting, objects that reappear in Sections 5.1 and 3: the minimal annihilator, the companion matrix and the spectral projectors.

Remark 2.9 (Setting and symbols).
- We consider an ODE $p(D)u = 0$ in one variable x, with $D = d/dx$ (in this section only; elsewhere D is a diffusion constant).
- The polynomial $p(\lambda) = \prod_j (\lambda - \lambda_j)^{m_j}$ has distinct roots $\lambda_j \in \mathbb{C}$ with multiplicities $m_j \geq 1$, so that $m = \deg p = \sum_j m_j$ is the *order* of the equation (not to be confused with the Sobolev index m of Section 3).
- The set of solutions is the *kernel* $\ker p(D)$, and

$$\ker p(D) = \Big\{ \sum_j q_j(x) e^{\lambda_j x} : \ \deg q_j < m_j \Big\},$$

where each q_j is a polynomial of degree $< m_j$.

Remark 2.10 (Why this form). For one root, $(D - \lambda)\big(q\, e^{\lambda x}\big) = q' e^{\lambda x}$: applying $D - \lambda$ lowers the degree of q by one. Hence $(D - \lambda_j)^{m_j}$ kills $q\, e^{\lambda_j x}$ exactly when $q^{(m_j)} = 0$, i.e. $\deg q < m_j$.

Each root therefore contributes the m_j functions $x^k e^{\lambda_j x}$ $(0 \leq k < m_j)$: these are the natural basis functions of the equation, finitely many, and together they form a basis of $\ker p(D)$, a space of dimension m.

Remark 2.11 (Example). $p(\lambda) = (\lambda - 1)^2(\lambda + 2)^3$ has $\lambda_1 = 1$, $m_1 = 2$, $\lambda_2 = -2$, $m_2 = 3$, order $m = 5$. A basis of the solutions is

$$e^x, \quad xe^x, \quad e^{-2x}, \quad xe^{-2x}, \quad x^2 e^{-2x},$$

so every solution is

$$c_1 e^x + c_2 x e^x + c_3 e^{-2x} + c_4 x e^{-2x} + c_5 x^2 e^{-2x}.$$

2.4 Identification by the Wronskian

Proposition 2.12 ((ODE identification)). *Let $u \in \ker p(D)$. The minimal annihilator of u has degree equal to the number of independent basis functions actually present in u; it equals p if and only if $u, u', \dots, u^{(m-1)}$ are linearly independent (nonvanishing Wronskian), i.e. iff every root of p appears in u with its full multiplicity.*

Reason. $\dim \operatorname{span}\{u, u', u'', \dots\}$ is the degree of the minimal annihilator (Lemma 2.7 in one variable); a missing root lowers it.

2.5 The companion matrix and spectral form

Write $p(\lambda) = \lambda^m + a_{m-1}\lambda^{m-1} + \cdots + a_0$ (monic) and put $X = (u, u', \dots, u^{(m-1)})^\top$. Then $X' = CX$ with

$$C = \begin{pmatrix} 0 & 1 & & \\ & 0 & \ddots & \\ & & \ddots & 1 \\ -a_0 & -a_1 & \cdots & -a_{m-1} \end{pmatrix}, \qquad X(x) = e^{xC} X(0).$$

The eigenvalues of C are the roots of p. For distinct roots the matrix exponential has the spectral form

$$\begin{aligned} e^{xC} &= \sum_{j=1}^{m} e^{\lambda_j x}\, \Pi_j, \\ \Pi_j &= \prod_{k \neq j} \frac{C - \lambda_k \mathrm{Id}}{\lambda_j - \lambda_k}, \end{aligned} \tag{4}$$

where the Π_j are the Lagrange projectors onto the eigenspaces. The last row of C carries the whole equation; C is a *matrix over the base field* for an ODE, but, as soon as the coefficients depend on x, or there is a second variable, its last row has entries that are operators and the object becomes a matrix over a noncommutative or polynomial ring. The invariant statement is that the whole equation corresponds to the cyclic module $\mathcal{R}/\operatorname{Ann}(u)$ (Section 2); the companion matrix is a way to compute with it relative to one chosen direction.

Proposition 2.13 ((Projectors of the companion matrix) (ledger II-T2) [proved]). *The projectors Π_j are explicit rational functions of the coefficients, with poles exactly where roots collide; they are polynomial in the coefficients if and only if the corresponding varieties are disjoint. They are* not *orthogonal in general: the companion matrix is not normal for $m = 2$ [false] (the contrary was an early conjecture).*

Hence Hilbert-space orthogonality arguments need an inner product adapted to the problem, not the standard one on the coefficient vector.

2.6 Tensor construction versus module construction

One may encode a PDE in a higher-order tensor of derivatives, $J = \partial_{x_{i_1}} \otimes \cdots \otimes \partial_{x_{i_n}}$, or by the companion form. The comparison, (ledger Gate 3): the *principal symbol tensor* $\sigma_m(P) \in \operatorname{Sym}^m(\mathbb{C}^2)$ only sees the highest-order part of P (for the heat operator it sees ξ^2 alone), so it is an invariant "at infinity"; the *cyclic module* $\mathcal{R}/\operatorname{Ann}(u)$ is the complete invariant. All subsequent results use the module.

Remark 2.14 (Takeaway). This section shows how an ODE's polynomial encodes its solution space. Its roots determine the exponential rates, their multiplicities determine the polynomial prefactors, the companion matrix represents differentiation, and spectral projectors isolate the different modes. The later sections generalize these ideas to multiple commuting derivatives, where the same structure survives but requires more algebraic machinery: the minimal annihilator becomes the ideal $\mathrm{Ann}(u)$ (Section 5.1) and the Lagrange projectors reappear as Eq. (11) (Section 3).

3 Groups are factors, not terms

In brief. The factor decomposition, its stability measured by an angle, and the regularity of the components.

3.1 Groups are factors, not terms

Suppose the operator is known and factored as

$$P = P_1^{e_1} \cdots P_N^{e_N}, \qquad Q_j = P_j^{e_j} \text{ pairwise coprime.} \tag{5}$$

Two ways of splitting a solution suggest themselves: one piece per *factor* Q_j, or one piece per additive *term* of the operator (for $u_t = \nu u_{xx} - c u_x$, one piece for diffusion and one for transport). Only the first works.

Theorem 3.1 (Factor decomposition)**.** *Let $\Omega \subset \mathbb{R}^2$ be bounded, connected and simply connected, and let P be as in* (5)*. Every $u \in \ker P(\partial)$ (in $\mathcal{D}'(\Omega)$ or $C^\infty(\Omega)$) can be written*

$$u = \sum_{j=1}^{N} u_j, \qquad Q_j(\partial)\, u_j = 0. \tag{6}$$

The tuples $(u_1, \ldots, u_N)$ that sum to zero form a space of exponential-polynomials of dimension exactly

$$\delta(P) = \sum_{i<j} \dim_{\mathbb{C}} \mathcal{R}/(Q_i, Q_j), \tag{7}$$

the number of intersection points of the varieties $V(Q_i)$, counted pair by pair and with multiplicity.

The number $\dim_{\mathbb{C}} \mathcal{R}/(Q_i, Q_j)$ therefore has two readings at once. Geometrically it counts the points where the curves $V(Q_i)$ and $V(Q_j)$ meet, with multiplicity. Analytically it is the dimension of the space of exponential-polynomials that solve both Q_i and Q_j, one exponential for each meeting point (times polynomials at a point of higher multiplicity). For two groups these are exactly the functions that can be moved from one group to the other without changing u: each meeting point of the two curves is one shared mode. For three or more groups $\delta(P)$ still gives the dimension of the ambiguity, but the ambiguous tuples need not be sums of modes shared by pairs.

In one variable the statement is Bézout's identity: if $aQ_1 + bQ_2 = 1$, then $u = bQ_2 u + aQ_1 u$ splits u into a solution of Q_1 and a solution of Q_2. In two variables "coprime" no longer gives $1 \in (Q_1, Q_2)$; it only gives finitely many common zeros, and the finite-dimensional quotient

$\mathcal{R}/(Q_1, Q_2)$ measures exactly how much the two groups overlap. The proof combines the exact sequence $0 \to \mathcal{R}/(Q_1Q_2) \to \mathcal{R}/(Q_1) \oplus \mathcal{R}/(Q_2) \to \mathcal{R}/(Q_1, Q_2) \to 0$ with the vanishing of the obstruction $\mathrm{Ext}^1(M, C^\infty(\Omega))$ for the finite-dimensional module $M = \mathcal{R}/(Q_1, Q_2)$, which reduces to $H^1_{dR}(\Omega) = 0$ (compare the fundamental principle of Ehrenpreis and Palamodov [5, 17]); see Appendix A.

Example 3.2 (Running example: two groups with a two-dimensional overlap). For $Q_1 = \tau + c\xi$ and $Q_2 = \tau - \nu\xi^2$ the common zeros are $(0, 0)$ and $(c^2/\nu, -c/\nu)$, both simple, so $\delta(P) = 2$. The ambiguity is spanned by the constant function and by $e^{(c^2/\nu)t-(c/\nu)x}$: a constant (or that growing exponential) can be moved from one group to the other without changing u. Among Fourier modes ($\xi = i\kappa$, $\kappa \in \mathbb{R}$) only the first point is visible: the $\kappa = 0$ mode belongs to both groups.

Projectors from the eigenvalues When each factor is an evolution factor, $Q_j = \tau - \Lambda_j(\xi)$ with pairwise distinct branches Λ_j, a component is an eigenfunction of ∂_t: $\partial_t u_j = \Lambda_j(\partial_x)u_j$. Applying $\prod_{k\neq j}(\partial_t - \Lambda_k(\partial_x))$ to $u = \sum_l u_l$ removes every u_l with $l \neq j$ and multiplies u_j by $\prod_{k\neq j}(\Lambda_j - \Lambda_k)(\partial_x)$. On a spatial Fourier mode κ at which no two branches meet this factor is a nonzero number, hence

$$\hat{u}_j = \prod_{k\neq j} \frac{\partial_t - \Lambda_k(i\kappa)}{\Lambda_j(i\kappa) - \Lambda_k(i\kappa)}\ \hat{u}(t, \kappa). \tag{8}$$

This is a multiplier, rational in κ, with poles exactly where two branches meet ($\Lambda_j(i\kappa) = \Lambda_k(i\kappa)$), that is, at the visible part of the overlap counted by $\delta(P)$. Formula (8) is the Lagrange interpolation formula for the spectral projectors of a matrix with distinct eigenvalues, applied to ∂_t restricted to one wavenumber: the *eigenvalues decide the groups*. For the running example, $\hat{u}_1(t, \kappa) = (\partial_t + \nu\kappa^2)\hat{u}/(\nu\kappa^2 - ic\kappa)$, which is singular only at $\kappa = 0$.

Why additive terms do not form groups

Proposition 3.3 (Terms do not give components)**.** *If $P = \mathcal{A}_1 + \mathcal{A}_2$ with constant coefficients, then* $\ker P \cap (\ker \mathcal{A}_1 + \ker \mathcal{A}_2) \subseteq \ker P \cap \ker(\mathcal{A}_1\mathcal{A}_2)$*, and the right-hand side is finite-dimensional whenever $V(\mathcal{A}_1) \cap V(\mathcal{A}_2)$ is finite.*

Proof. If $u = a + b$ with $\mathcal{A}_1 a = 0$ and $\mathcal{A}_2 b = 0$, then $\mathcal{A}_1\mathcal{A}_2 u = \mathcal{A}_2\mathcal{A}_1 a + \mathcal{A}_1\mathcal{A}_2 b = 0$ because the operators commute. At a common zero of P and $\mathcal{A}_1\mathcal{A}_2$ either $\mathcal{A}_1 = 0$, and then $\mathcal{A}_2 = P - \mathcal{A}_1 = 0$, or $\mathcal{A}_2 = 0$ and then $\mathcal{A}_1 = 0$; so $V(P, \mathcal{A}_1\mathcal{A}_2) = V(\mathcal{A}_1) \cap V(\mathcal{A}_2)$. When this set is finite, the ideal $(P, \mathcal{A}_1\mathcal{A}_2)$ has finite codimension, its common kernel consists of exponential-polynomials with exponents in that set (Lemma 2.7), and it is finite-dimensional. □

For the heat equation, $\mathcal{A}_1 = \partial_t$ and $\mathcal{A}_2 = -\nu\partial_x^2$: every sum $f(x) + a(t) + b(t)x$ of a solution of $\mathcal{A}_1$ and one of $\mathcal{A}_2$ is killed by $\mathcal{A}_1\mathcal{A}_2 \sim \partial_t\partial_x^2$, while a heat mode $e^{-\nu\kappa^2 t}\cos\kappa x$ is not. A generic heat solution is *not* "a diffusion part plus a stationary part". The leftover of any term-wise split is the interaction term of Section 4.1.

Where the decomposition fails The hypotheses of Theorem 3.1 are not technical. On a disconnected domain different pieces can satisfy different equations. On a domain with a hole the decomposition can fail: the Laplacian factors as $(\partial_x + i\partial_y)(\partial_x - i\partial_y)$, yet $\log|z|^2$ is

harmonic on an annulus and is not a sum of a holomorphic and an antiholomorphic single-valued function. The obstruction is the first de Rham cohomology of Ω, which is why the algorithm works on rectangles.

3.2 How well separated are the groups? The Friedrichs angle

Theorem 3.1 is algebraic: it says that the split exists, not that it is stable. Stability is a geometric property of the component spaces $M_j = \ker_{H^s} Q_j$, the solutions of each factor in the Sobolev space $H^s(\Omega)$. Let

$$\mathcal{S}:\ \textstyle\bigoplus_{j=1}^{N} M_j \to H^s(\Omega), \qquad \mathcal{S}(u_1,\dots,u_N) = \sum_j u_j,$$

be the *sum operator*, with $\|(u_1,\dots,u_N)\|^2 = \sum_j \|u_j\|_{H^s}^2$, and define the *conditioning constant*

$$\gamma_N = \inf_{0 \neq v \perp \ker \mathcal{S}} \frac{\|\mathcal{S}v\|^2}{\|v\|^2} = \inf \sigma\big(\mathcal{S}^*\mathcal{S}\big|_{(\ker \mathcal{S})^\perp}\big). \tag{9}$$

It is the worst case, over all tuples of pieces, of the size of the sum relative to the size of the pieces. Tuples in $\ker \mathcal{S}$ (the overlap of Theorem 3.1) are excluded. By the triangle and Cauchy–Schwarz inequalities, $\|\sum_j u_j\|^2 \le \big(\sum_j \|u_j\|\big)^2 \le N \sum_j \|u_j\|^2$, hence

$$0 \le \gamma_N \le N.$$

If the spaces are mutually orthogonal, $\|\sum_j u_j\|^2 = \sum_j \|u_j\|^2$ and $\gamma_N = 1$. At the other end, if M_2 approaches M_1 one can take $u_2 \approx -u_1$: the pieces stay of unit size while their sum becomes arbitrarily small, so $\gamma_N \to 0$. A small γ_N is the signature of groups that can almost cancel each other.

For two groups, $\gamma_2 = 1 - c_F$, where the *Friedrichs cosine*

$$c_F = \sup\big\{|\langle a, b\rangle| :\ a \in M_1 \ominus (M_1 \cap M_2),\ b \in M_2 \ominus (M_1 \cap M_2),\ \|a\| = \|b\| = 1\big\} = \cos\theta_F$$

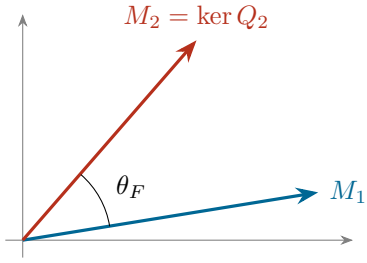
is the cosine of the smallest nonzero angle θ_F between the two spaces (Fig. 3) [2]. In regression terms, M_1 and M_2 are two blocks of regressors and c_F is the largest correlation between a combination of the first block and one of the second: the closer the blocks are to collinear, the less the data can attribute to each.

Theorem 3.4 (Stability of the split)**.** *Every $u \in \ker_{H^s} P$ has components $u_j \in H^s$ if and only if $\sum_j M_j$ is closed and dense in $\ker_{H^s} P$. Closedness is equivalent to $\gamma_N > 0$, and then*

$$\Big(\sum_j \|u_j\|_{H^s}^2\Big)^{1/2} \le \gamma_N^{-1/2}\, \|u\|_{H^s} \tag{10}$$

for the minimum-norm components. If the component spaces are finite-dimensional, as for the sampled kernels used by the algorithm, the sum is always closed and density holds whenever the sampled kernel of P is spanned by those of its factors, so stability reduces to $\gamma_N > 0$.

Density in H^s is a genuine condition on infinite-dimensional spaces. On convex domains solutions of $P(\partial)u = 0$ are limits of exponential solutions, uniformly on compact subsets [14, 10], and every exponential solution lies in $\sum_j M_j$; whether this extends to the H^s norm up to the boundary for general P we do not claim.



$c_F = \cos\theta_F, \quad \gamma_2 = 1 - c_F$
error amplification $\gamma_2^{-1/2}$
running example: $\theta_F \approx 0.60$, $\gamma_2 \approx 0.17$

Figure 3: The Friedrichs angle θ_F between the solution spaces of two factors. A small angle means that large pieces can almost cancel, so recovering the pieces from their sum amplifies errors by up to $\gamma_2^{-1/2} = (1-\cos\theta_F)^{-1/2}$. The values quoted are those measured on the running example in the companion algorithm paper.

Two warnings come with the theorem. First, $\gamma_N > 0$ can fail on infinite-dimensional component spaces: the pieces of a smooth u may then be one derivative rougher than u itself. Second, for $N \geq 3$ groups the pairwise cosines do not determine γ_N: two pairs can be well separated while the three spaces are nearly dependent, so the full Gram operator of $\mathcal{S}$ must be used.

3.3 An explicit projector for evolution factors

For an evolution operator with $P = \prod_{j=1}^{N}(\tau - a_j(\xi))$ and distinct a_j, a component is an eigenfunction of ∂_t: $\partial_t u_j = a_j(\partial_x)u_j$. Applying $\prod_{k\neq j}(\partial_t - a_k(\partial_x))$ to $u = \sum_l u_l$ kills every u_l with $l \neq j$ (it contains the factor $\partial_t - a_l$) and multiplies u_j by $\prod_{k\neq j}(a_j - a_k)(\partial_x)$. Hence

$$u_j = \prod_{k\neq j} \frac{\partial_t - a_k(\partial_x)}{a_j(\partial_x) - a_k(\partial_x)}\, u, \tag{11}$$

the Lagrange formula of Eq. (4) with C replaced by ∂_t. The result is a Fourier multiplier in x that is rational in ξ, with poles where $a_j(\xi) = a_k(\xi)$; this matches the characterization "projectors rational in ξ iff the factors are coprime" (ledger II-T2/T3).

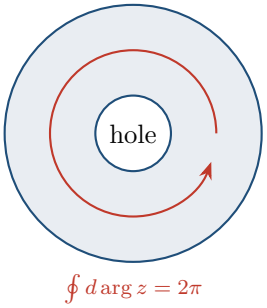
3.4 Where decomposition fails

Proposition 3.5 ((Holes) (ledger C12-P10) [proved]). *On a domain with a hole, the decomposition fails for some P.*

Figure 4 shows the standard obstruction. In the theorem the hypothesis "simply connected" means $H^1_{dR}(\Omega) = 0$, exactly what removes this obstruction; a cohomological criterion for non-simply connected Ω is also available (ledger OP-FUN, C12-Q6).

3.5 Regularity: when are the components in H^m?

The decomposition is an algebraic statement; analysis enters when we want components in a Sobolev space. For two components the governing condition is geometric: the two component spaces M_1 and M_2 must be separated by a positive angle (Figure 5); the next theorem makes this precise for any number of components. Two difficulties appear along the way; they are collected in the box after the theorem.



$u = \log|z-\alpha|^2$ solves $\Delta u = 0$ on the annulus,
but $u \neq h(z) + \overline{g(z)}$ with h, g
single-valued holomorphic: the logarithm
has monodromy (winding number $\neq 0$).

Figure 4: The Laplacian $\partial_x^2 + \partial_y^2 = (\partial_x + i\partial_y)(\partial_x - i\partial_y)$ factors, but the factor decomposition fails on a punctured disc: the obstruction is topological ($H^1_{dR} \neq 0$).

Proposition 3.6 ((Loss of a derivative) (ledger C12-C1) [false])**.** *"If $u \in H^m(\Omega)$ then all $u_j \in H^m$" is false: one derivative can be lost. In other words, the pieces of a smooth function may be rougher than the function itself.*

Theorem 3.7 ((Regularity criterion) (ledger C12-T3, C12-P7) [proved])**.** *All components lie in H^m for every $u \in \ker_{H^m} P$ iff the sum $\sum_j \ker_{H^m} Q_j$ is* closed *and* dense *in $\ker_{H^m} P$. Closedness is equivalent to $\gamma_N > 0$, where*

$$\gamma_N = \inf \sigma\big(S^*S\big|_{(\ker S)^\perp}\big),$$
$$S : \bigoplus_{j=1}^{N} \ker Q_j \to H^m, \qquad S(u_1, \dots, u_N) = \sum_{j=1}^{N} u_j.$$

For $N = 2$ this is the condition $c_F < 1$ on the Friedrichs cosine. Density holds on every bounded convex Ω, for every P and m (ledger OPF5B-T1) [proved mod. Hörmander [9], Stein].

Remark 3.8 (Reading γ_N). • The space $\bigoplus_j \ker Q_j$ consists of the N-tuples $v = (u_1, \dots, u_N)$ with norm $\|v\|^2 = \sum_j \|u_j\|^2_{H^m}$;
- $Sv = \sum_j u_j$ is their sum;
- S^* is the adjoint of S;
- $(\ker S)^\perp$ removes the tuples that sum to zero (the ambiguity of Theorem 3.1);
- $\inf \sigma$ is the bottom of the spectrum.

Since S^*S restricted to $(\ker S)^\perp$ is self-adjoint and nonnegative, the bottom of its spectrum is the minimum of the Rayleigh quotient

$$\frac{\langle S^*Sv, v\rangle}{\|v\|^2} = \frac{\|Sv\|^2}{\|v\|^2}.$$

Hence the formula of the theorem is equivalent to the more intuitive

$$\boxed{\gamma_N = \inf_{\substack{v \in (\ker S)^\perp \\ v \neq 0}} \frac{\|Sv\|^2_{H^m}}{\|v\|^2}}$$

Remark 3.9 (In words). γ_N is the worst case, over all tuples of pieces, of the ratio between the size of the sum and the size of the pieces.

- If $\gamma_N > 0$ the pieces cannot nearly cancel each other, and

$$\|v\| \le \gamma_N^{-1/2} \|Sv\| :$$

  the pieces are controlled by the sum, so recovering them from u is stable.
- If $\gamma_N \to 0$ there are large pieces whose sum is tiny, and the recovery is unstable.

The case $N = 2$. One finds $\gamma_2 = 1 - c_F$, where

$$c_F = \sup\{|\langle a, b\rangle| : \ a \in M_1 \ominus (M_1 \cap M_2),\ b \in M_2 \ominus (M_1 \cap M_2),\ \|a\| = \|b\| = 1\}$$

is the Friedrichs cosine: the part of v lying in $M_1 \cap M_2$ contributes the ratio 2 and does not affect the infimum, and on the rest

$$\|a + b\|^2 \ge (1 - c_F)(\|a\|^2 + \|b\|^2)$$

with near-equality attained. Hence $c_F < 1 \iff \gamma_2 > 0$, as stated.

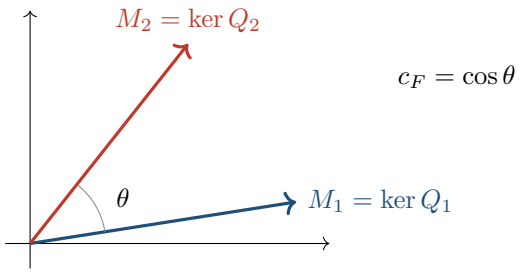


Figure 5: Friedrichs angle θ between two component spaces $M_1 = \ker Q_1$ and $M_2 = \ker Q_2$. A larger angle means a better-conditioned recovery of the pieces from their sum.

For $N = 2$, Figure 5 reads as follows: $c_F = \cos\theta$; if $\theta > 0$ the sum $M_1 + M_2$ is closed and the pieces depend continuously on u; as $\theta \to 0$ the split becomes unstable. For $N \ge 3$ the pairwise angles do not suffice; γ_N, the smallest nonzero eigenvalue of S^*S, is the right conditioning number: it is the conditioning of the linear map "components $\mapsto$ sum", exactly like the smallest nonzero singular value in a least-squares problem.

Remark 3.10 (Warning). [title=Two problems with regularity] **Problem 1: stability [false].** Algebraically the split always exists (Theorem 3.1), but the pieces may be less regular than u. When the angle between the component spaces tends to 0 (that is, $\gamma_N \to 0$) the map "$u \mapsto$ pieces" becomes unstable, and one derivative can be lost (Proposition 3.6). For the algorithm: check $\gamma_N > 0$ before using the components in H^m (Step 5).

Problem 2: three or more components [open]. For $N \ge 3$ the pairwise cosines $c_F(i, j)$ do not decide whether the sum of kernels is closed: an abstract Hilbert-space counterexample exists (ledger S-CHECK-4), and whether it can occur for differential operators is open (ledger C12-Q2).

Remark 3.11 (Why it matters). Algebraically the split always exists (Theorem 3.1); analytically, the components are well behaved exactly when the component spaces are separated by a positive angle (Theorem 3.7). The next section stops asking for a split into factors and asks what survives when one splits by additive terms instead.

4 Inside one group: the interaction term

In brief. What remains when a factor is built from several terms: an exact formula, its cost, its instability, and what happens when the terms do not commute.

4.1 Inside one group: the interaction term and its energy

A single factor may have several additive terms: $Q_2 = \tau - \nu\xi^2$ has one, but the advection–diffusion factor $\tau + c\xi - \nu\xi^2$ has two. Proposition 3.3 says that such a factor cannot be split further into one solution per term. We now measure exactly what is left over.

Consider an evolution $u_t = \mathcal{A}u$ with $\mathcal{A} = \mathcal{A}_1 + \cdots + \mathcal{A}_K$ (the terms), the flow $\Phi(t) = e^{t\mathcal{A}}$ and the flows of the single terms $\Phi_n(t) = e^{t\mathcal{A}_n}$. If term n acted alone it would change the datum by $\Delta_n u_0 = (\Phi_n(t) - \mathrm{Id})u_0$. The *interaction term* is the part of the evolution that is not the sum of these single-term changes,

$$E_{\mathrm{int}}(t) = u(t) - u_0 - \sum_{n=1}^{K} \big(\Phi_n(t)u_0 - u_0\big). \tag{12}$$

Theorem 4.1 (Interaction term)**.** *If the terms commute (in particular, for constant coefficients),*

$$E_{\mathrm{int}}(t) = \Big[\prod_{n=1}^{K}(\mathrm{Id} + \Delta_n) - \mathrm{Id} - \sum_{n=1}^{K}\Delta_n\Big]u_0 = \sum_{\mathcal{E}\subseteq\{1,\dots,K\},\ |\mathcal{E}|\geq 2} \Big(\prod_{n\in\mathcal{E}}\Delta_n\Big)u_0.$$

For two terms $E_{\mathrm{int}} = \Delta_1\Delta_2 u_0$. On Fourier modes, with symbols $\hat{\mathcal{A}}_n(\kappa)$ of the terms,

$$\widehat{E_{\mathrm{int}}}(t,\kappa) = \Big[\prod_{n} e^{t\hat{\mathcal{A}}_n(\kappa)} - 1 - \sum_{n}\big(e^{t\hat{\mathcal{A}}_n(\kappa)} - 1\big)\Big]\hat{u}_0(\kappa). \tag{13}$$

Proof. Commuting terms give $\Phi = \prod_n \Phi_n = \prod_n(\mathrm{Id}+\Delta_n)$. Expanding the product over subsets $\mathcal{E} \subseteq \{1,\dots,K\}$ and removing the subsets with zero or one element gives the formula; on Fourier modes every operator is multiplication by a number. □

Formula (13) costs $O(K)$ operations per wavenumber, although writing out all cross terms would need $2^K - K - 1$ of them. For advection–diffusion, $|\widehat{E_{\mathrm{int}}}|/|\hat{u}_0| = |(e^{-ic\kappa t}-1)(e^{-\nu\kappa^2 t}-1)| \leq 2$: the interaction is of the same size as the solution, so it is not a small correction that a regularizer could remove.

Proposition 4.2 (No stable removal)**.** *Let $Q = \tau + c\xi - \nu\xi^2$ (advection–diffusion, terms $-c\partial_x$ and $\nu\partial_x^2$) on a bounded rectangle $\Omega = (0,T)\times(a,b)$. For every $s \geq 0$,*

$$\sup\Big\{\|E_{\mathrm{int}}\|_{L^2(\Omega)}/\|u\|_{H^s(\Omega)} :\ u \in \mathrm{EP},\ Q(\partial)u = 0,\ u \neq 0\Big\} = \infty,$$

so $u \mapsto E_{\mathrm{int}}$ has no continuous extension to $\ker_{H^s} Q$. Moreover, a solution of $Q(\partial)u = 0$ is a sum $v + w$ of solutions of the single-term evolutions $(\partial_t + c\partial_x)v = 0$ and $(\partial_t - \nu\partial_x^2)w = 0$ only if it lies in a finite-dimensional space of exponential-polynomials.

The test solutions are $e^{\lambda_\zeta t+\zeta x}$ with real $\zeta > c/\nu$ and $\lambda_\zeta = \nu\zeta^2 - c\zeta$, legitimate data on a bounded interval: diffusion alone amplifies them by $e^{\nu\zeta^2 t}$, the solution grows only like $e^{\lambda_\zeta t}$, and the interaction must cancel a difference that grows faster than any power of ζ (Appendix A). On a periodic domain with bounded wavenumbers the Fourier multiplier above is bounded by 2; the instability comes from the non-periodic window, which is the situation of real data. The second statement means that a model which writes the data as a sum of single-term pieces, with any penalty on the pieces, leaves a remainder on generic data, and that remainder is the interaction.

Energy We measure the size of a group and of the interaction by L^2 energies on the observation window. The *energy share* of group j and the *interaction ratio* are

$$w_j = \frac{\|u_j\|}{\sum_k \|u_k\|}, \qquad \rho = \frac{\|E_{\text{int}}(T)\|}{\|u(T) - u_0\|}, \tag{14}$$

where T is the length of the time window. The shares w_j say how much of the data each group explains; ρ says how far the evolution inside a multi-term factor is from the superposition of its terms ($\rho = 0$ for a factor with a single term). Interactions occur only *inside* a factor: different factors superpose exactly, which is another way of saying that factors, not terms, are the right groups.

4.2 Definition via the flows of the single terms

Let $S(t) = e^{tA}$ and $S_n(t) = e^{tA_n}$. The contribution that term n *alone* would make to the solution is $S_n(t)u_0 - u_0$. The remainder is the interaction term

$$E_{\text{int}}^{\text{add}}(t) = u(t) - u_0 - \sum_{n=1}^{N} \big(S_n(t)u_0 - u_0\big). \tag{15}$$

(The superscript "add" records that this is the *additive* definition; below we write E_{int} for $E_{\text{int}}^{\text{add}}$ unless the distinction matters.)

4.3 Exact formula in the commuting case

Remark 4.3 (Why constant coefficients commute, and what Δ_n is). Constant-coefficient operators are polynomials in ∂_t, ∂_x (Section 2) and the ring $\mathcal{R}$ is commutative, so any two terms commute: $A_1A_2 = A_2A_1$. For commuting operators

$$e^{t(A_1+A_2)} = e^{tA_1}e^{tA_2}, \qquad \text{hence} \qquad S = \prod_n S_n;$$

without commutativity this fails (Section 4.6).

We call

$$\Delta_n = S_n(t) - \mathrm{Id}$$

the *increment* produced by term n alone, where Id is the identity operator and t is fixed; thus $S_n = \mathrm{Id} + \Delta_n$, and $\Delta_n u_0 = S_n(t)u_0 - u_0$ is exactly the contribution of term n in (15). Products of operators are compositions.

Remark 4.4 (Fourier variables). For constant coefficients it is convenient to pass to Fourier variables. With

$$\hat{u}(\kappa) = \int u(x) e^{-i\kappa x}\, \mathrm{d}x,$$

where $\kappa \in \mathbb{R}$ is the frequency:

- the derivative ∂_x becomes multiplication by $i\kappa$;
- each term A_n becomes multiplication by its *symbol* $a_n(\kappa)$ (a polynomial in $i\kappa$);
- e^{tA_n} becomes multiplication by the number $e^{ta_n(\kappa)}$.

Operators turn into numbers and compositions into products of numbers, so for a fixed κ every computation below is scalar.

Theorem 4.5 ((Interaction term) (ledger P1-T2, P1-T3) [proved])**.** *Assume the A_n commute (e.g. constant coefficients). Write $\Delta_n = S_n(t) - \mathrm{Id}$. Then*

$$E_{\mathrm{int}}^{\mathrm{add}} = \Big[\prod_{n=1}^{N} (\mathrm{Id} + \Delta_n) - \mathrm{Id} - \sum_{n=1}^{N} \Delta_n \Big] u_0. \tag{16}$$

For $N = 2$ this is simply $E_{\mathrm{int}} = \Delta_1 \Delta_2 u_0 = (S_1 - \mathrm{Id})(S_2 - \mathrm{Id})u_0$. In Fourier variables, with $\hat{A}_n = a_n(\kappa)$,

$$\widehat{E_{\mathrm{int}}}(t, \kappa) = \Big[\prod_{n} \big(e^{ta_n(\kappa)}\big) - 1 - \sum_{n} \big(e^{ta_n(\kappa)} - 1\big) \Big] \hat{u}_0(\kappa).$$

Proof. Commutation gives $S = e^{t\sum A_n} = \prod_n S_n$. Substitute $S_n = \mathrm{Id} + \Delta_n$ in (15). For $N = 2$, expand the product:

$$\begin{aligned} S_1 S_2 &= (\mathrm{Id} + \Delta_1)(\mathrm{Id} + \Delta_2) \\ &= \mathrm{Id} + \Delta_1 + \Delta_2 + \Delta_1 \Delta_2, \end{aligned}$$

and $\sum_n (S_n - \mathrm{Id}) = \Delta_1 + \Delta_2$, so the remainder in (15) is $\Delta_1 \Delta_2 u_0$.

For general N, the Δ_n commute, so

$$\prod_{n} (\mathrm{Id} + \Delta_n) = \sum_{S \subseteq \{1, \dots, N\}} \prod_{n \in S} \Delta_n;$$

removing the subsets with $|S| = 0$ and $|S| = 1$ leaves

$$E_{\mathrm{int}} = \sum_{|S| \geq 2} \Delta_S\, u_0, \qquad \Delta_S = \prod_{n \in S} \Delta_n.$$

In words: the interaction is the sum of products of at least two increments, and appears only when at least two terms act together. □

A concrete case. Take advection–diffusion, $u_t = -cu_x + Du_{xx}$, with Gaussian datum $u_0 = g(x; \sigma^2)$, where $g(x; v) = e^{-x^2/2v}/\sqrt{2\pi v}$. The two flows act as a translation and a Gaussian smoothing:

$$S_1 u_0 = g(x - ct; \sigma^2), \qquad S_2 u_0 = g(x; s^2), \qquad u = g(x - ct; s^2), \qquad s^2 = \sigma^2 + 2Dt.$$

Hence

$$E_{\mathrm{int}} = g(x - ct; s^2) - g(x - ct; \sigma^2) - g(x; s^2) + g(x; \sigma^2).$$

Figure 6(a) plots it: it is comparable in size to u itself, so it cannot be discarded as a small correction. In Fourier variables the ratio $|\widehat{E_{\mathrm{int}}}|/|\hat{u}_0| = |(e^{-ic\kappa t} - 1)(e^{-D\kappa^2 t} - 1)|$ is at most 2; it vanishes at $\kappa = 0$ and whenever $c\kappa t \in 2\pi\mathbb{Z}$ (the translation returns to the identity), see Figure 6(b).

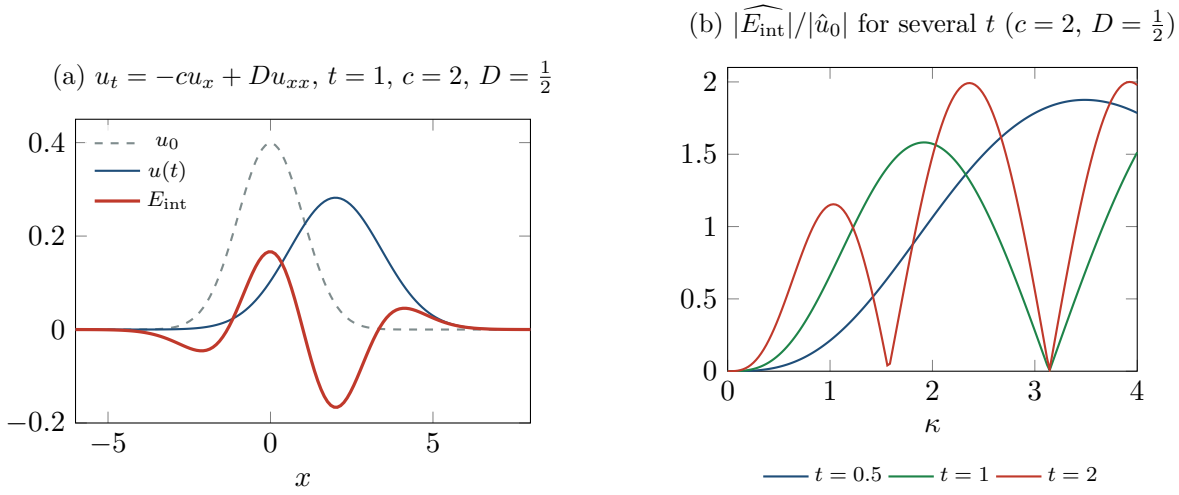


Figure 6: Advection–diffusion with Gaussian datum ($\sigma^2 = 1$, $c = 2$, $D = \frac{1}{2}$). (a) u_0, $u(t)$ and E_{int} at $t = 1$. (b) The Fourier ratio $|\widehat{E_{\text{int}}}|/|\hat{u}_0|$ for three times.

4.4 Cost: $O(N)$ versus 2^N

Expanding (16) in the Δ_n produces one monomial per subset of $\{1, \dots, N\}$ with at least two elements: $2^N - N - 1$ cross terms (Figure 7: the red nodes are the cross terms). Writing the *complete table* of cross terms is therefore exponential. Evaluating a *value* is cheap, because we never expand the product. The algorithm uses exactly the product (Fourier) form of Theorem 4.5,

$$\widehat{E_{\text{int}}}(t, \kappa) = \Big[\prod_{n=1}^{N} e^{t a_n(\kappa)} - 1 - \sum_{n=1}^{N} \big(e^{t a_n(\kappa)} - 1\big) \Big] \hat{u}_0(\kappa),$$

which for a fixed κ needs the N numbers $e^{t a_n(\kappa)}$, $N - 1$ products and N subtractions: $O(N)$ operations once $\hat{u}_0(\kappa)$ is known.

Remark 4.6 (Two remarks on cost). **What "value" means.** Here "value" means one Fourier value; recovering $E_{\text{int}}(t, x)$ at a point of space needs an inverse Fourier transform, whose cost is not counted in the $O(N)$.

Where 2^N comes from. The count $2^N - N - 1$ refers only to writing out all cross terms one by one; the product form never does so (Step 6 of the algorithm).

Proposition 4.7 ((Complexity) (ledger P1-T3) [proved]). *In the commuting exponential-type class, a value of E_{int} costs $O(M(N + \sum d))$ for M evaluation points; aggregated by order $O(MN^2)$; the full table $\Theta(M2^N(L + 1)^2)$.*

4.5 Time-dependent perturbation theory

Remark 4.8 (Why this subsection). The product formula (16) needs commuting terms. When terms do not commute, or when boundary conditions are imposed, there is no closed formula, but E_{int} is still defined by (15). Perturbation theory expands the flow in powers of the interaction. From this expansion one reads

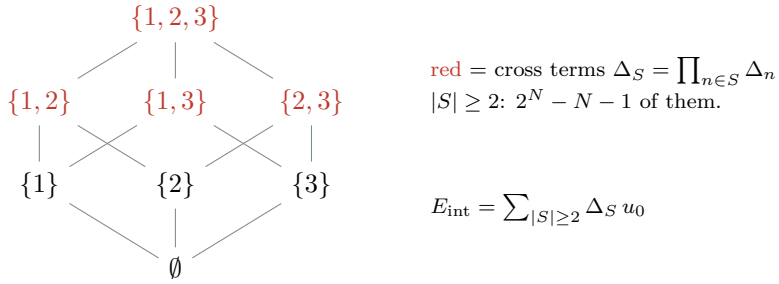


Figure 7: Subset lattice for $N = 3$ terms: one node per subset S of terms, one monomial Δ_S per node with $|S| \geq 2$. The count $2^N - N - 1$ is what makes the full table exponential.

(i) that E_{int} is well defined for all times (convergence),
(ii) approximations obtained by truncating the series order by order, and
(iii) the size of E_{int} for small t.

Split $A = A_1 + B$, where A_1 is the principal term (for instance diffusion) and B, the remaining terms, is treated as a perturbation of A_1. The Dyson–Phillips series is

$$\begin{aligned} S(t) &= \sum_{k\geq 0} S^{(k)}(t), \\ S^{(0)}(t) &= e^{tA_1}, \\ S^{(k+1)}(t) &= \int_0^t e^{(t-s)A_1}\, B\, S^{(k)}(s)\, \mathrm{d}s, \end{aligned} \tag{17}$$

the same expansion that gives the interaction picture of quantum mechanics [3].

Remark 4.9 (Reading the series). • $S^{(0)} = e^{tA_1}$ is the flow of A_1 alone;
• $S^{(k)}$ collects the contributions in which B acts k times, at intermediate times $s \in (0, t)$, with the flow of A_1 in between.

The terms with $k \geq 1$ are therefore the part of the flow in which B acts on the evolution generated by A_1, that is, the interaction. (When A_1 and B commute, the first-order term is

$$\int_0^t e^{(t-s)A_1} B e^{sA_1}\, \mathrm{d}s = t\, B e^{tA_1},$$

consistent with Section 4.)

What it gives. The series converges for all t when A_1 generates an even-order dissipative semigroup (e.g. diffusion) and B is a lower-order L^∞ perturbation, also with Dirichlet conditions (ledger P3-T1) [proved mod. Miyadera–Voigt [6]]: so E_{int} is well defined and can be approximated by truncation even without a closed formula. Each $S^{(k)}$ is the order-k part of the interaction. For small times, for boundary-induced interaction, with k the order of contact of the datum with the boundary, $\|E_{\text{int}}\| \sim t^{k/2+3/4}$ (ledger P3-T3, OP-F5A) [proved]: the smoother the datum is against the boundary (larger k), the faster the interaction becomes negligible as $t \to 0$.

4.6 Non-commuting terms

If $[A_1, A_2] \neq 0$, then $e^{t(A_1+A_2)} \neq e^{tA_1} e^{tA_2}$; by Baker–Campbell–Hausdorff,

$$e^{tA_1} e^{tA_2} = \exp\Big(t(A_1 + A_2) + \tfrac{t^2}{2}[A_1, A_2] + O(t^3)\Big).$$

Proposition 4.10 ((Lie-algebra dichotomy) (ledger P1-T4, P1-T5) [proved]). *If the Lie algebra generated by the terms is solvable, an exact formula with finitely many scalar functions exists (Wei–Norman). Otherwise only a local expansion exists, and* ***no finite product of the flows of the individual terms*** *reproduces* $e^{t(X+\varepsilon Y)}$.

Remark 4.11 (Why it matters). For commuting terms the interaction is explicit and cheap to evaluate, but it cannot be made to vanish nor be computed stably in H^m; for non-commuting terms even the explicit form is lost in general. These are the linear-case limits against which the nonlinear extension of Section 6 must be read.

5 When does a solution determine its equation?

In brief. The identification criterion, uniqueness, and what finite data can and cannot certify. This explains why the algorithm of the companion paper never reports "identified".

5.1 When does a solution determine its equation?

Let Ω be open and connected (on a disconnected set the pieces may obey different equations), let $u \in \mathcal{D}'(\Omega)$ satisfy $P(\partial)u = 0$, and let $P = \prod_i P_i^{e_i}$ (up to a nonzero constant factor) be the factorization of P into pairwise non-associate irreducible polynomials P_i with multiplicities e_i. The question is whether $\mathrm{Ann}(u) = (P)$: then every constant-coefficient equation satisfied by u is a consequence of P, and P is determined by u.

Theorem 5.1 (Identification criterion). *Put* $v_i = (P/P_i)(\partial)\, u$ *and* $v_i' = (P/P_i^{e_i})(\partial)\, u$.

(a) $\mathrm{Ann}(u) = (P)$ *if and only if* $v_i \notin \mathrm{EP}(\Omega)$ *for every* i.

(b) $V(\mathrm{Ann}\, u) = V(P)$ *if and only if* $v_i' \notin \mathrm{EP}(\Omega)$ *for every* i.

(c) If P *is square-free, (a) and (b) coincide. If* P *is monic in* τ *of degree* m, *then* $\mathrm{Ann}(u) = (P)$ *if and only if the only* $(\varphi_0, \dots, \varphi_{m-1}) \in \mathbb{C}[\xi]^m$ *with* $\sum_{j<m} \varphi_j(\partial_x)\, \partial_t^j u = 0$ *is zero.*

The idea is to strip one irreducible factor and look at what is left. The function v_i solves $P_i(\partial)v_i = 0$. If v_i is a generic solution of that factor, no smaller operator kills u. If v_i is an exponential-polynomial, some h coprime to P_i kills it, and then $h \cdot P/P_i$ is an annihilator of u that is not a multiple of P. The converse uses Lemma 2.4; the full proof is in Appendix A.

Proposition 5.2 (Uniqueness and the intrinsic operator). *(a) If* $\mathrm{Ann}(u) = (P)$, *then* P *is determined by* u *up to a nonzero constant and is the unique annihilator of minimal degree.*

(b) $\mathrm{Ann}(u) = (P)$ *if and only if* $\Sigma_{\mathrm{p}}(u) = \emptyset$ *and* $G_u \sim P$ *(equality up to a constant).*

(c) For any annihilator P, *the multiplicity of* P_i *in* G_u *equals* $e_i - \max\{k \le e_i : (P/P_i^k)(\partial)u \in \mathrm{EP}\}$. *Hence* G_u *is computed from any annihilator by at most* $\sum_i e_i$ *membership tests "*$\in \mathrm{EP}$*".*

Example 5.3 (The heat equation, continued). For $P = \tau - \nu\xi^2$ (irreducible) the criterion reads: $\mathrm{Ann}(u) = (P)$ if and only if $u \notin \mathrm{EP}$. One mode $e^{-\nu\kappa^2 t + i\kappa x}$ has $\mathrm{Ann}(u) = (\tau + \nu\kappa^2,\ \xi - i\kappa)$, a point of $V(P)$; it cannot distinguish the heat equation from the pure decay $\partial_t + \nu\kappa^2$, or from any other equation whose variety passes through the point $(-\nu\kappa^2, i\kappa)$. The heat kernel is not in EP, so every constant-coefficient equation it satisfies is a multiple of $\partial_t - \nu\partial_x^2$.

Remark 5.4 (Curves and points). The two cases of Example 5.3 sit at the two ends of the decomposition (3). A finite sum of modes $u \in \mathrm{EP}$ has an annihilator with finitely many zeros, so

$$u \in \mathrm{EP}: \qquad G_u = 1, \qquad \Sigma_{\mathrm{p}}(u) = \{(\tau_j, \xi_j)\} \text{ finite (its exponents)},$$

whereas the zero set of a nonconstant polynomial in two variables is a whole algebraic curve. For the heat kernel the situation is reversed:

$$u = t^{-1/2} e^{-x^2/(4\nu t)}: \qquad G_u = \tau - \nu\xi^2, \quad V(G_u) \text{ a curve}, \qquad \Sigma_{\mathrm{p}}(u) = \emptyset.$$

By Proposition 5.2(b) it is exactly this pair, an empty point spectrum together with $G_u \sim P$, that gives $\mathrm{Ann}(u) = (P)$: the solution fills the whole curve of its equation and contains no isolated mode besides.

5.2 What finite data can and cannot certify

Theorem 5.1 speaks about u, not about finitely many numbers. Two facts frame what an algorithm can promise.

Recall that $\mathcal{R}u = \mathrm{span}\{\partial_t^j \partial_x^k u : \ j, k \geq 0\}$ is the vector space generated by the derivatives of u (Section 2.3). The *jet* of order M of u at a point $z_0 = (t_0, x_0)$ is the list of its derivatives of order at most M at that point; for example

$$j^2_{z_0}(u) = (u, \ u_t, \ u_x, \ u_{tt}, \ u_{tx}, \ u_{xx})\big|_{(t,x)=z_0}.$$

Proposition 5.5 (Finite data)**.** *Let Ω be connected and $z_0 \in \Omega$.*

(i) If $\dim_{\mathbb{C}} \mathcal{R}u \leq n$, the jet of order $2n-1$ of u at z_0 determines $\mathrm{Ann}(u)$ among all functions u' with $\dim_{\mathbb{C}} \mathcal{R}u' \leq n$.

(ii) For every u and every M there is a polynomial, hence an exponential-polynomial, with the same jet of order M at z_0; and for finitely many sample points there is an exponential-polynomial with the same values. Consequently, without a bound on $\dim \mathcal{R}u$, no finite set of jets or samples proves $\mathrm{Ann}(u) = (P)$.

Proof. (ii) The Taylor polynomial of order M at z_0 has the same jet, and polynomial interpolation in two variables matches any finite set of values. Both are exponential-polynomials, and an exponential-polynomial never has $\mathrm{Ann} = (P)$ when P has an irreducible factor (Theorem 5.1: all v_i are then exponential-polynomials). (i) The space $\mathcal{R}u$ is invariant under differentiation and, by Lemma 2.7, consists of real-analytic exponential-polynomials. Let K_M be the subspace of elements of $\mathcal{R}u$ whose jet of order M vanishes at z_0. Then $K_{M+1} = \{f \in K_0 : \ \partial_t f, \partial_x f \in K_M\}$, so once $K_{M+1} = K_M$ the chain is constant, and a constant K_M is a derivative-invariant space of functions vanishing at z_0 with all their derivatives, hence $\{0\}$. Since $\dim K_0 \leq n-1$, the chain reaches $\{0\}$ by $M = n-1$: the jet of order $n-1$ at z_0 is injective on $\mathcal{R}u$, and likewise on $\mathcal{R}u'$. For Q of degree at most n, $Q \in \mathrm{Ann}(u)$ iff the jet of order $n-1$ of $Q(\partial)u$ vanishes at z_0, a condition on the jet of order $2n-1$ of u; the same holds for u'. Finally $\mathrm{Ann}(u)$ has codimension at most n, so a Gröbner basis for a degree-compatible order consists of polynomials of degree at most n; the same holds for $\mathrm{Ann}(u')$, and two ideals generated in degree at most n that contain the same polynomials of degree at most n are equal. □

The consequence for learning is a design rule: without an a priori bound on $\dim \mathcal{R}u$, which data never supply, an algorithm can refute a candidate equation, or report that it is compatible with the data at the resolution available, but it cannot certify it. Its output must carry a label that says which of these happened.

6 Nonlinear equations

In brief. The same two questions, asked of Burgers-type equations: the interaction term for the flows of single terms, behaviour before, at and after a shock, and identification relative to a dictionary.

Starting from: a scalar equation in one space variable, $u_t = \sum_n T_n(u)$, each T_n linear or polynomial in the jet $(u, u_x, u_{xx}, \dots)$; Ω bounded and simply connected; the exact solution is known. *Why:* to test which of the notions of Sections 5.1–4 survive when the terms are nonlinear. *Role:* extension of the chain. *Result:* weaker, relative statements (exact expansion of E_{int}, sharp asymptotics across a shock for Burgers, identification relative to a dictionary). Benchmarks: Burgers, KdV–Burgers, Fisher–KPP; KdV and NLS only as controls.

We consider $u_t = \sum_n T_n(u)$ with each T_n linear or polynomial in the jet $(u, u_x, u_{xx}, \dots)$. Benchmarks: Burgers $u_t = -uu_x + \nu u_{xx}$, KdV–Burgers, Fisher–KPP; integrable cases (KdV, NLS) serve as controls.

6.1 Interaction term for flows of single terms

With S_n the (nonlinear) flow of $u_t = T_n(u)$ alone, (15) still defines $E_{\text{int}}^{\text{add}}$. Its Taylor expansion in t:

Theorem 6.1 ((Jet expansion) (ledger S4T-T1) [proved]). $E_{\text{int}}^{\text{add}} = \dfrac{t^2}{2} \displaystyle\sum_{n \neq m} T_n'[\, T_m(u_0)\,] + O(t^3)$,

where $T_n'[v]$ is the Fréchet derivative of T_n at u_0 in the direction v.

Example 6.2 (Burgers). $T_1 = -uu_x$, $T_2 = \nu u_{xx}$. Then $T_1'[v] = -(vu_x + uv_x)$ and $T_2'[v] = \nu v_{xx}$, so

$$T_1'[T_2 u_0] = -\nu\,(u_0'' u_0' + u_0 u_0'''),$$
$$T_2'[T_1 u_0] = -\nu\,(3u_0' u_0'' + u_0 u_0'''),$$

and half the sum times t^2 gives

$$\begin{aligned} E_{\text{int}} &= \tfrac{1}{2}(T_1'[T_2 u_0] + T_2'[T_1 u_0])t^2 + O(t^3) \\ &= -\nu(u_0 u_0''' + 2u_0' u_0'')t^2 + O(t^3). \end{aligned}$$

6.2 Before, at, and after the shock

The interaction term of Burgers changes qualitatively with time, and each regime calls for different tools:

Before shock formation smoothness permits perturbative expansions and derivative-based estimates.

At breaking the inviscid gradient blows up, invalidating estimates that require smoothness.

After shock formation pointwise behavior can be discontinuous, so L^1 estimates become more informative than pointwise bounds.

Figure 8 recalls why a shock appears at all in the flow of the single term $-uu_x$: along each characteristic u is constant and the characteristic moves with speed u, so faster parts overtake slower ones. For $u_0 = \sin x$ the characteristics are

$$x = \xi + t \sin \xi$$

(the parameter s in the figure plays the role of the time t); since

$$\frac{\partial x}{\partial \xi} = 1 + t\cos\xi$$

vanishes first at $\xi = \pi$, the breaking time is $t^* = 1$.

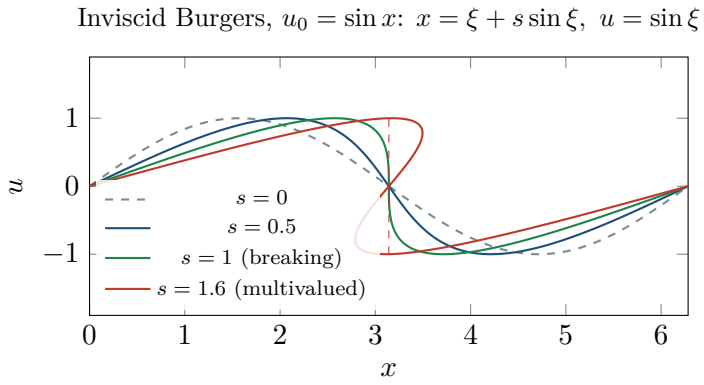


Figure 8: Characteristic solution of the inviscid flow of $-uu_x$. At $s = 1$ the profile has a vertical tangent: the gradient blows up. Beyond that the curve folds; the physical (entropy) solution replaces the fold by a jump at $x = \pi$ (equal-area rule), shown dashed. This breaking is what makes E_{int} discontinuous after the shock.

Before the shock ($t < t^*$): smooth regime. Smoothness permits perturbative expansions and derivative-based estimates.

- E_{int} has a closed form at first order in ν, with remainder $O(\nu^2\varepsilon^2 t^3)$ (ledger S4T-T2) [proved]; it blows up at the breaking time t^* (ledger S4T-P3). (This is the same family of expansions as Theorem 6.1 and Example 6.2.)
- The Cole–Hopf substitution $u = -2\nu\varphi_x/\varphi$ turns Burgers into the heat equation $\varphi_t = \nu\varphi_{xx}$; the inverse scattering transform linearizes KdV and NLS. These are used as the "exact linearization" of the nonlinear solution [8].

At breaking ($t = t^*$): estimates that need smoothness fail. The inviscid gradient u_x blows up (vertical tangent, $s = 1$ in Figure 8), so every statement that needs bounds on $u_x, u_{xx}, \ldots$ stops being usable at t^*: the closed form of E_{int} blows up, and estimates that require smoothness, such as those behind the jet expansion of Theorem 6.1, are no longer valid.

After the shock ($t > t^*$): discontinuous regime. The flow of $-uu_x$ alone forms a shock, so E_{int} jumps by amounts independent of ν; this is forced by the definition. A function with a jump is not in H^1 in one dimension, so the Sobolev norms of Sections 3 and 4 are no longer the natural ones: pointwise values are unreliable, while the *measure* of the discrepancy is well

behaved. The meaningful quantities are therefore L^1 sizes, collected below:

$$\|E_{\mathrm{int}}\|_{L^1} = \Theta(\nu), \tag{18}$$

$$\sup_{t\le T} \|E_{\mathrm{int}}\|_{L^1} = \nu|\log\nu| + O(\nu), \tag{19}$$

$$\|S - S_1 S_2\|_{L^1} = \Theta(\sqrt{\nu}). \tag{20}$$

The first line holds at fixed t, each shock contributing $4\nu\log(2\cosh(\rho/2))$; the second has exact leading coefficient 1. Sources: the first line is (ledger OPA-T1) [proved mod. Kružkov uniqueness [12]]; the second is (ledger OPF5A-T1/T2); the third is (ledger OPF5D-T6) [proved]. The second line holds for data with a single birth of shock such as $\sin x$, and is conditional on a localization lemma in general [conditional].

Regime	What happens	Tools and results
Before the shock, $t < t^*$	the profile is smooth, characteristics do not cross	jet expansion (Thm. 6.1), first-order closed form of E_{int}, Cole–Hopf
At breaking, $t = t^*$	the inviscid gradient blows up (vertical tangent)	smoothness-based estimates fail; E_{int} blows up
After the shock, $t > t^*$	the single-term flow is discontinuous (equal-area rule)	L^1 estimates: $\Theta(\nu)$, $\nu\|\log\nu\|$, $\Theta(\sqrt{\nu})$

Table 2: The three regimes of the Burgers interaction term and the tools that apply in each.

6.3 Identification relative to a dictionary

Nonlinear terms cannot be handled by the ideal $\mathrm{Ann}(u)$ of Section 2; instead one fixes a finite candidate list. Choose a dictionary $\mathcal{D}$ of monomials of degree $\le d$ in the jet, of order $\le k$, and ask which polynomial relations $F(u, u_x, \dots) = 0$ the data satisfy. Notation: V is the span of $\mathcal{D}$, $I(u)$ the ideal of all differential relations satisfied by u, $[F]$ the differential ideal generated by F, and $\mathcal{J}_u$ the jet-level obstruction (a different object from the cofactor J_u of Section 2, hence the calligraphic letter).

Theorem 6.3 ((Dictionary criterion) (ledger S4ID-T1) [proved]). *For smooth u on any Ω there is an exact sequence $0 \to [F]\cap V \to I(u)\cap V \to \mathcal{J}_u \cap \psi(V) \to 0$. In particular*

$$I(u)\cap V = [F]\cap V \iff \mathcal{J}_u \cap \psi(V) = 0.$$

The correct statement is therefore "the relations are exactly F *modulo its differential consequences*"; the literal "$I(u)\cap V = \langle F\rangle$" is false in complete dictionaries (they contain $\partial_x F$, etc.) (ledger S4ID-P1). Concrete criteria: for Burgers with N shocks, u identifies the equation in the evolutionary dictionary iff $d \le N$ (ledger S4ID-T2, OP-T7); for N-soliton KdV the identifying polynomial has total degree $(N+2)2^{N-1}$ (generic $N = 2$: 8) (ledger OP-K15) [proved mod. Novikov 1974 [15]]. Finally, "closed form $\Rightarrow$ does not identify" is itself a theorem (ledger S4ID-T4).

6.4 Linear versus nonlinear: the same idea, two structures

Sections 2–4 and the present one answer the same question: *what does a single solution u remember about its equation?* In both settings we collect all relations satisfied by u and ask

	Linear approach (Sections 2–4)	Nonlinear approach (this section)
Space of candidate equations	the ring $\mathcal{R} = \mathbb{C}[\tau, \xi]$ of constant-coefficient operators	a finite *dictionary* $\mathcal{D}$ of jet monomials of degree $\leq d$ and order $\leq k$; $V = \operatorname{span} \mathcal{D}$
Relations satisfied by u	$\operatorname{Ann}(u)$, an ideal of $\mathcal{R}$	$I(u)$, the ideal of all differential relations
What the equation generates	the principal ideal (P)	the differential ideal $[F]$
Criterion for identification	$\operatorname{Ann}(u) = (P)$ (Theorem 5.1)	$I(u) \cap V = [F] \cap V$, i.e. F *modulo its differential consequences* (Theorem 6.3)
What escapes the equation	the cofactor J_u: the finitely many exponentials in u ($\sigma_p(u) = V(J_u)$)	the jet-level obstruction $\mathcal{J}_u \cap \psi(V)$ (exact sequence of Theorem 6.3)
Solutions that do not identify	$u \in \mathrm{EP}$	closed-form solutions ((ledger S4ID-T4))
Concrete criteria	finite linear algebra (Lemma 2.7)	Burgers with N shocks identifies iff $d \leq N$; N-soliton KdV: degree $(N+2)2^{N-1}$

Table 3: The linear and nonlinear identification problems side by side.

when these relations are generated by the equation we were given. What changes is the algebraic structure in which the relations live.

Remark 6.4 (Where the parallel breaks)*.* **Linear case.** The solutions form a vector space and the annihilators an ideal: one can add, multiply by operators and factor, which is why the Nullstellensatz applies and the test is a finite-dimensional computation.

Nonlinear case. There is no linear structure on the solutions: one must fix a finite space V of candidates in advance, and the identification is *relative to* V, never absolute. The literal statement

$$I(u) \cap V = \langle F \rangle$$

is false in complete dictionaries, which contain $\partial_x F$, so the correct statement is "modulo differential consequences".

Remark 6.5 (Takeaway)*.* Both approaches ask what one solution determines about its equation. The linear one answers with the annihilator ideal $\operatorname{Ann}(u)$ and its generator G_u; the nonlinear one answers with a dictionary and the differential ideal of relations, and only up to differential consequences. In the algorithm of the companion paper Steps 1–6 use the structure of $\operatorname{Ann}(u)$, while Step 7 switches to the dictionary and returns a relative identification.

Part II: beyond constant coefficients (Phases 6–10)

Sections 7–13 extend the theory of Sections 2–6 (Phases 0–5) to systems, variable coefficients, noisy and finite data, encoders, dictionaries and the extended algorithm (Phases 6–10; Phase 11 was the audit of the whole). Each result names its proof as "Dim. An" or "Dim. Bn" in the companion *Catalogue of proofs* (directory `paper/analytical_complete`).

7 Systems of equations and gauge

Starting from: a vector field $u = (u_1, \dots, u_k)$ on a connected open set $\Omega \subset \mathbb{R}^n$ that solves a system $P(\partial)u = 0$ with a *matrix* of constant coefficients $P \in R^{l\times k}$, $R = \mathbb{C}[\zeta_1, \dots, \zeta_n]$. The operator P is not given. *Why:* the equations of physics that are not scalar (Dirac, Maxwell, elasticity) are systems; Phases 1–5 treated one scalar equation. We ask the same three questions: can one solution determine the system, when can a sum of kernels be split, and what does a change of gauge do to what can be known? *Role:* extends Sections 5–3 from R to matrices over R. It supplies Step 3b of the extended algorithm (Section 13). The proofs are Dim. A32–A35 and B73–B78.

7.1 From one equation to a module

For a scalar equation the annihilator $\mathrm{Ann}(u)$ was an ideal of R. For a system the object that replaces it is a *module*. The rows of P generate a submodule $U_P \subseteq R^{1\times k}$, and the quotient

$$M_P = R^{1\times k}/U_P$$

is the module of the system: solutions of P are exactly the homomorphisms $M_P \to \mathcal{A}$ (Malgrange–Oberst duality). A vector u *identifies* P when its annihilator $\mathrm{Ann}(u) = \{r : r(\partial)u = 0\}$ equals U_P, i.e. when the map $M_P \to \mathcal{A}$, $[r] \mapsto r(\partial)u$, is injective.

Remark 7.1 (Intuition). Think of M_P as the list of everything the system says, up to consequences. A single function u must carry all of it in an injective way. The obstruction is local: at a point λ the solutions of the system that are pure exponentials $e^{\lambda\cdot x}$ form one line, but M_P may contain a plane of "independent" behaviours at that point. A plane cannot be embedded in a line.

7.2 When does an identifying solution exist?

Define the *socle* of M at a maximal ideal $\mathfrak{m}_\lambda$ as the set of elements of M killed by $\mathfrak{m}_\lambda$; it is the space of exponential behaviours $e^{\lambda\cdot x}$ the module contains.

Theorem 7.2 (Existence of an identifying solution (ledger SYS-EX) [proved])**.** *Let $\Omega \subseteq \mathbb{R}^n$ be non-empty, open and connected. A finitely generated module M embeds in $C^\infty(\Omega)$ (or $\mathcal{D}'(\Omega)$) if and only if $\dim_{\mathbb{C}} \mathrm{soc}_{\mathfrak{m}}(M) \le 1$ for every maximal ideal $\mathfrak{m}$. In particular a system P has an identifying solution if and only if $\dim \mathrm{Hom}_R(R/\mathfrak{m}, M_P) \le 1$ for every maximal $\mathfrak{m}$.* [proof: Dim. SYS-EX]

Example 7.3. Two ODE systems For $n = 1$ let $P = \mathrm{diag}(s-1, s-1)$ (two copies of $u' = u$). The socle at $s = 1$ has dimension 2: no function u ever identifies it. For $P = \begin{pmatrix} s-1 & 1 \\ 0 & s-2 \end{pmatrix}$ all socles have dimension ≤ 1, and $u = (ae^x - be^{2x}, be^{2x})$ with $ab \neq 0$ identifies it. In general an ODE system of rank r with invariant factors $e_1 \mid \cdots \mid e_r$ has an identifying solution iff $e_1 = \cdots = e_{r-1} = 1$ (ledger SYS-T0) (known).

Theorem 7.4 (Square non-singular systems in two or more variables (ledger SYS-C1) [proved])**.** *If $n \ge 2$ and $P \in R^{k\times k}$ has $\det P \neq 0$, every socle of M_P is zero, so on every connected Ω some $u \in \mathrm{Sol}(P)$ identifies P. The associated primes of M_P are the prime factors of $\det P$.* [proof: Dim. SYS-C1]

Dirac in $3+1$ dimensions, the matrix Klein–Gordon operator and the Lamé system of elasticity are all of this type. They always have an identifying field.

Remark 7.5 (Warning). [title=Where this is not new] For $\Omega = \mathbb{R}^n$ the existence criterion is known (Oberst, Pommaret, Macaulay: the number of trajectories needed to generate a system). The new point here is only that Ω may be any connected open set, including one where $C^\infty(\Omega)$ is not an injective module.

7.3 Sums of kernels for systems

For a scalar $P = Q_1 Q_2$ we proved $\ker P = \ker Q_1 + \ker Q_2$ on a simply connected region (Theorem 3.1). For systems the analogue is stated with the *least common left multiple*: if U_1, U_2 are the row modules of two sub-systems and $N = R^{1\times k}/(U_1 + U_2)$, then $\mathrm{Sol}(Q_1) + \mathrm{Sol}(Q_2) = \mathrm{Sol}(P)$ requires $U_1 \cap U_2 = U_P$.

Theorem 7.6 (Sum of kernels (ledger SYS-S1) [proved])**.** *On every open Ω, $\mathrm{Sol}(Q_1) + \mathrm{Sol}(Q_2) = \mathrm{Sol}(P)$ implies $U_1 \cap U_2 = U_P$. Conversely, if $U_1 \cap U_2 = U_P$ and $\mathrm{Ext}^1(N, \mathcal{A}) = 0$ the equality holds; this is the case when Ω is convex, and when N has finite length and $H^1_{dR}(\Omega) = 0$. The intersection $\mathrm{Sol}(Q_1) \cap \mathrm{Sol}(Q_2)$ is finite-dimensional iff N has finite length, with dimension $\delta_{\mathrm{sys}} = \ell(N)$. For one scalar equation in two variables this is Theorem 3.1.* [proof: Dim. SYS-S1]

Remark 7.7 (Counterexample: Simple connectivity does not suffice for infinite-length quotients). [false] There are a simply connected open $\Omega \subset \mathbb{R}^2$ (a "U"-shaped domain) and $Q_1 = (1, 0)$, $Q_2 = (1, \partial_x)$ with $U_1 \cap U_2 = U_P$ but $\mathrm{Sol}(Q_1) + \mathrm{Sol}(Q_2) \subsetneq \mathrm{Sol}(P)$, in C^∞ and in $\mathcal{D}'$. The obstruction is the failure of P-convexity of Ω, not its topology. [proof: Dim. SYS-S1]

7.4 Dirac in Hamiltonian form

Write the free or perturbed Dirac equation as $L = I\partial_t + \sum_k \alpha^k \partial_k + B$ with arbitrary constant 4×4 matrices α^k, B (no Clifford relations are assumed).

Theorem 7.8 (Dirac coefficients are read exactly (ledger H-DIRAC) [proved])**.** *If $\psi \in \mathrm{Sol}(L_0)$ identifies L_0 and L is another operator of this form with $L\psi = 0$, then $L = L_0$: the matrices α^k and B are determined by ψ exactly, with no ambiguity. An identifying ψ always exists. For massive Dirac in $3+1$ dimensions $M_D \cong (R/\mathfrak{q})^2$ with $\mathfrak{q} = (p_0^2 - |p|^2 - m^2)$, and ψ identifies D iff ψ_3, ψ_4 are independent over $R/\mathfrak{q}$; in $1+1$ dimensions ψ identifies D_2 iff $\psi_2 \notin \mathrm{EP}$.* [proof: Dim. H-DIRAC]

Example 7.9. Where the Dirac coefficients come from Normalising the coefficient of ∂_t to the identity removes the ambiguity $D \sim VD$ of the covariant form $\gamma^\mu \partial_\mu$. In $1+1$ dimensions the identification test is just the one of Section 5: the second component must not be an exponential-polynomial.

7.5 Gauge, magnetic fields, and Lamé

Proposition 7.10 (Gauge covariance and what invariant data reveal (ledger G-L0, G-T2) [proved])**.** *Under $A \mapsto A + d\chi$, $u \mapsto e^{i\chi}u$ the covariant operators transform covariantly, so the identification tests are gauge-invariant. If the observation is invariant under the phase, a reconstruction of the potential A recovers at most its gauge class $[A]$: $A' \sim A$ iff $dA' = dA$ and the holonomies agree (Aharonov–Bohm). On a simply connected region, $A' \sim A$ iff $dA' = dA$.* [proof: Dim. G-T2]

Proposition 7.11 (Magnetic Schrödinger: one field does not identify (A, V) (ledger SYS-N3) [proved]). *For $n \geq 2$ and a nowhere-zero solution u the pairs (A', V') compatible with u form an infinite-dimensional affine space, parametrised by divergence-free fields k; their magnetic fields differ, so they are not gauge-equivalent, even in the non-stationary case. The obstruction is a count: one complex scalar equation against $n+1$ real unknowns.* [proof: Dim. SYS-N3]

Proposition 7.12 (Lamé in space (ledger LAME-3) [proved]). *For $\rho, \mu \neq 0$, $\lambda+\mu \neq 0$, $\lambda+2\mu \neq 0$, the system has $\det P = q_P q_S^2$ with Bass multiplicities 1 and 2; an identifying displacement field exists, and u identifies it iff $\operatorname{div} u$ satisfies no equation besides the wave equation of speed $\sqrt{(\lambda+2\mu)/\rho}$ and the two transverse polarisations of $\operatorname{rot} u$ satisfy no extra relation. A pure P-wave ($\operatorname{rot} u = 0$) never identifies.* [proof: Dim. LAME-3]

Proposition 7.13 (Dirac $1+1$ as an evolution: no uniform product (ledger SYS-E1, E2, E3) [proved]). *Writing $\psi_t = (X + mY)\psi$ with X the transport and Y the mass term, the operators $X, \operatorname{ad}_X^k Y$ are linearly independent, so no finite product of transport and mass flows with ξ-independent times equals $e^{t(X+mY)}$ uniformly in the mass. The Lie algebra they generate is infinite-dimensional. An explicit product (Euler angles) exists, with times that are not polynomial in ξ and with non-local Fourier multipliers.* [proof: Dim. SYS-E]

Proposition 7.14 (Systems from finite data (ledger SYS-R1) [proved]). *From exponential-polynomial data one can identify only systems with finite-dimensional solution space; for an ODE system the identifying solution exists iff $e_1 = \cdots = e_{r-1} = 1$.* [proof: Dim. SYS-R1]

Remark 7.15 (Why it matters). For a system the unknown is a module and the datum is one vector function. The only obstruction to identification is the socle, a purely algebraic and local count, and it vanishes for every non-singular square system in $n \geq 2$ variables. What a single field can never do is separate gauge from physics.

8 Variable coefficients

Starting from: a rectangle $\Omega = I \times J \subset \mathbb{R}^2$ and a solution u of $iu_t = -u_{xx} + V\,u$, i.e. $L_0 u = Vu$ with $L_0 = i\partial_t + \partial_x^2$, where the potential $V(x)$ or $V(t,x)$ is unknown. Solutions are *strong* (class SF): $u \in C^{1,2}(\Omega)$ and the equation holds almost everywhere. *Why:* in physics the coefficients are rarely constant: a Schrödinger particle feels a potential. Phases 1–5 identified constant coefficients. Here the unknown is a function, and the first question is whether a single solution can reveal it at all. *Role:* extends Section 5 from $\mathbb{C}$ to function coefficients. It supplies Step 3c of the extended algorithm (Section 13). The proofs are Dim. A36–A40 and B78.

8.1 Too much freedom: a huge class identifies nothing

Suppose we allow coefficients in any algebra $\mathcal{K}$ of functions closed under differentiation (all smooth functions, all analytic functions). Then even one solution that never vanishes carries no information beyond itself.

Theorem 8.1 (Rank-one annihilator (ledger VAR-N1, VAR-T1(a)) [proved]). *Let $u \in C^\infty(\Omega)$ have no zeros, with $a = u_t/u$ and $b = u_x/u$ in $\mathcal{K}$. Then the annihilator of u among differential operators with coefficients in $\mathcal{K}$ is generated by $\partial_t - a$ and $\partial_x - b$, and the module $\mathcal{K}[\partial]u$ is isomorphic to $\mathcal{K}$ itself. For an affine family $L_\theta = L_0 + \sum \theta_k T_k$ the compatible parameters are $\theta_0 + \ker \Phi_u$, so θ is identified iff $\Phi_u(\theta) = \sum \theta_k T_k u$ is injective.* [proof: Dim. VAR-N1]

Remark 8.2 (Intuition). If the coefficient class is closed under differentiation, a zero-free u can write each of its own derivatives as "a coefficient times u", so every law is a law about u itself and nothing more. To identify anything the class of coefficients must be restricted in advance.

Remark 8.3 (Warning). [title=Zeros matter] With zeros the statement fails: $u = t - x^2$ near 0 satisfies no law $\partial_t + \alpha\partial_x + \beta$ with smooth coefficients, because at the origin it would give $1 = 0$.

8.2 A potential $V(x)$ is read from any non-zero solution

For a time-independent potential the situation is the opposite: nothing can hide.

Theorem 8.4 (Identification of $V(x)$ (ledger VAR-T1(b)) [proved])**.** *Let $u \not\equiv 0$ be a strong solution with potential $V \in L^\infty_{\mathrm{loc}}(J)$, and $X_0 = \{x : u(t,x) = 0\ \forall t\}$. Then X_0 is closed and discrete; if $L_0 u = W(x)u$ for another W, then $W = V$ almost everywhere; and*

$$V(x) = \frac{iu_t + u_{xx}}{u} \qquad \text{at every } (t,x) \text{ with } u(t,x) \neq 0.$$

[proof: Dim. VAR-T1b]

Remark 8.5 (Intuition). If u vanished along infinitely many vertical lines accumulating at x_0, cutting u off to the right of x_0 would give another strong solution that vanishes on a half-strip. Schrödinger waves have infinite speed, and a Carleman estimate (a weighted energy inequality) shows that a solution vanishing on a strip vanishes everywhere. So u cannot have such lines, and the formula $V = (iu_t + u_{xx})/u$ holds on a set of full measure.

Example 8.6. The harmonic oscillator For $V = x^2$ and the Hermite functions $h_0, \dots, h_3$ the set X_0 is empty or $\{0\}$, and $(iu_t + u_{xx})/u$ returns x^2 exactly. The same formula recovers the Pöschl–Teller and Airy potentials.

8.3 A potential $V(t,x)$: silent strips

Theorem 8.7 (Time-dependent potentials (ledger VAR-T1(c)) [proved])**.** *For $V \in C(\Omega)$, the potentials compatible with u are $V + \{W : W = 0$ on $\Omega \setminus Z(u)\}$, where $Z(u) = \{u = 0\}$; so V is identified iff* $\operatorname{int} Z(u) = \emptyset$. *If* $\operatorname{int} Z(u) \neq \emptyset$ *there is an open interval $T' \subset I$ with $u = 0$ on $T' \times J$ (a* silent strip*). Silent strips occur: Tychonoff's series gives a solution vanishing for $t \leq 0$ and non-zero for $t > 0$. Hence "every $u \not\equiv 0$ identifies $V(t,x)$" is false. [proof: Dim. VAR-T1c]*

Remark 8.8 (Counterexample: Data that switch on from the boundary). [false] On a bounded domain without boundary conditions, a solution can be identically zero up to a time and non-zero afterwards (Tychonoff's series). While it is silent every potential supported in the silent time is compatible with it. The failure is in the time direction: unique continuation holds in x but not in t.

8.4 Stability and the phase

Theorem 8.9 (Stability of the reconstruction (ledger VAR-P3) [proved])**.** *On a rectangle K with $|u_1| \geq \delta > 0$: $\|V_1 - V_2\|_\infty \leq \delta^{-1}(1 + \|V_2\|_\infty)\|u_1 - u_2\|_{C^{1,2}(K)}$; the map is not continuous from C^1 to L^∞ (an explicit oscillation); and under an a priori bound M on $\|e_{tt}\|_\infty, \|e_{xxx}\|_\infty$ one has a Hölder modulus with sharp exponent $1/3$. It is a modulus of continuity, not a certificate.*
[proof: Dim. VAR-P3]

Theorem 8.10 (Unobserved phase: V modulo constants (ledger VAR-P4) [proved]). *If only the class $\{e^{i\theta(t)}u\}$ is observed, the compatible potentials are exactly $\{V + c : c \in \mathbb{R}\}$. For $P(\partial_t, L_q)u = 0$ with P unknown, (P, q) is defined only modulo $q \to q + c$, $P(\tau, \ell) \to P(\tau, \ell - c)$: a redundancy of the parametrisation.* [proof: Dim. VAR-P4]

Proposition 8.11 (Dirac $1+1$ with a potential (ledger VAR-P5) [proved]). *Finite speed of propagation leaves regions that the data never visit: for $m = V = 0$ a travelling bump $\psi = f(x-t)v_+$ vanishes for $x \le 0.5$ and $x \ge 1.7$, so every potential supported there is compatible. The matrix potential is determined at x_0 iff $\operatorname{span}_t\{\psi(t, x_0)\} = \mathbb{C}^2$.* [proof: Dim. VAR-P5]

8.5 Other results of this phase

- **Algebra of differential operators with polynomial coefficients** (Weyl algebra): for a D-finite u the annihilator is a principal left ideal and u identifies iff its derivatives are independent over K; Airy is irreducible and every solution identifies; the oscillator is reducible exactly for $E \in \pm(2\mathbb{N} + 1)$ (ledger VAR-T2a, VAR-T6, VAR-E1, VAR-E2).
- **Dynamical Lie algebras**: $\dim \mathfrak{g}_V < \infty$ iff $V \in \operatorname{span}\{1, x, x^2\}$; the modified Pöschl–Teller potential has an explicit evolution through a Darboux chain (ledger VAR-P8).
- **Non-autonomous splitting**: $E^{\text{mult}} = \iint [A_2(s_1), A_1(s_2)] + O(t^3)$; commuting at equal times is not enough (ledger VAR-P9). Dyson–Phillips converges for bounded time-dependent perturbations (ledger VAR-P11); Floquet logarithms are bounded but not differential (ledger VAR-P12).

These are list cards of the catalogue (Part IV); the closures made in Phase 10 are in Section 12.

Remark 8.12 (Why it matters). A function-valued coefficient is identified by one solution exactly when the solution cannot hide: for $V(x)$ it cannot (infinite speed and unique continuation); for $V(t, x)$ it can fall silent; for a huge class of coefficients, nothing is identified. Stability is of order two and the phase costs one constant.

9 Convergence, noise and the limits of learning

Starting from: one analytic function u on a connected open set $\Omega \subset \mathbb{R}^n$, observed through its Taylor jet at a point p_0 (the datum at step K is $j_K u$), or through a noisy copy $\tilde{u}$ with $\|\tilde{u} - u\| \le \delta$ in a stated norm. *Why:* Sections 5 and 5.2 proved that no finite amount of data certifies identification. A real experiment gives more and more data: does the identified operator converge, how fast, and what can a noisy measurement still say? *Role:* supplies Steps 1–4 and 7 of the extended algorithm (Section 13). The proofs are Dim. A41–A43.

9.1 Identification in the limit

At step K the rule computes, for each degree $m \le K$, the space $N_{m,K}$ of operators of degree $\le m$ whose action on u has vanishing Taylor coefficients up to order $K - m$. These spaces decrease with K and their limit is $\operatorname{Ann}(u) \cap R_{\le m}$. The *minimal-degree rule* outputs the gcd of $N_{m^*(K),K}$ at the smallest degree $m^*(K)$ where it is non-zero.

Theorem 9.1 (Identification in the limit (ledger CONV-L1, CONV-T1) [proved]). *If $\operatorname{Ann}(u) \neq 0$ with minimal degree d, the rule outputs $g_d = \gcd(\operatorname{Ann}(u) \cap R_{\le d})$ from some step on; if $\operatorname{Ann}(u) = (P)$*

it outputs P, without knowing $\deg P$*; if* $\mathrm{Ann}(u) = 0$ *the degree* $m^*(K) \to \infty$*. The sequence of outputs takes at most* $(d+1)(d+2)/2$ *values. But:*

- *the step K^* after which the output is correct is not bounded by any function of d;*
- *no function of the jet of order K is a correct certificate: any finite jet is shared by a polynomial, whose annihilator is a different ideal.*

[proof: Dim. CONV-T1]

Example 9.2. Convergence at an unpredictable time For $u = 1+x^{N+2}e^{x^2}$ one has $\mathrm{Ann}(u) = (\tau)$. The rule outputs "no law of degree 1" for $K \le N+1$ and τ from $K = N+2$, for every N: nothing in the first $N+1$ derivatives announces the law.

9.2 No learner decides whether a law exists

Theorem 9.3 (Analytic Gold theorem (ledger CONV-T2, CONV-T2′) [proved])**.** *For any sequence of maps φ_K from jets of order $\le K$ to $\{law, no\ law\}$ (no computability or measurability assumed) there is a real entire function u on which $\varphi_K(j_K u)$ does not converge to the correct answer. The same holds restricted to the class where* $\mathrm{Ann}(u)$ *is principal or zero; hence no learner identifies* $\mathrm{Ann}(u)$ *on that class. [proof: Dim. CONV-T2]*

Remark 9.4 (Intuition). Build u as a limit of polynomials. Whenever the learner says "law" on the current polynomial (it must, eventually), add Taylor coefficients, chosen outside a proper algebraic set, that destroy every law of degree $\le k$. The limit function is differentially transcendental yet agrees with a polynomial where the learner said "law", infinitely often.

Remark 9.5 (Takeaway). "$\mathrm{Ann}(u) \neq 0$" is *verifiable in the limit* (a rule that is eventually right when it holds) but not *decidable in the limit.*

9.3 Noisy data

Theorem 9.6 (Noise: discontinuity, conditional stability, discrepancy rule (ledger CONV-T3, T4, T5) [proved])**.** *(T3) In spaces where polynomials are dense and $e^{|z|^2}$ lies, the sets* $\{\mathrm{Ann} = 0\}$ *and* $\{\mathrm{Ann} \neq 0\}$ *are both dense: $u \mapsto \mathrm{Ann}(u)$ is discontinuous everywhere. (T4) At a fixed level n_0 with a margin $\rho = \sigma_2(\Phi(u)) > 0$, the minimal singular vector of $\Phi(\tilde{u})$ is within* $\sin\angle \le 2\delta/\rho$ *of the true law, an order that is optimal; under an L^2 error and an a priori H^{n_0+s} bound the rate is Hölder, and $\sigma_1(\Phi(\tilde{u})) > \delta$ certifies a misfit. (T5) The discrepancy rule $m(\delta) = \min\{m : \sigma_1(\Phi_m(\tilde{u})) \le \delta_m\}$ returns the correct degree d and converges at rate δ once $\delta < \delta_0 = \frac{1}{2}\min_{m<d} s_m$; if* $\mathrm{Ann}(u) = 0$*, $m(\delta) \to \infty$. The threshold δ_0 cannot be bounded below from the data. [proof: Dim. CONV-T4]*

Example 9.7. The heat kernel from noisy data For $u = (t+1)^{-1/2}e^{-x^2/(4(t+1))}$ (minimal degree 2, $\mathrm{Ann} = (\tau - \xi^2)$), the rule returned $m(\delta) = 2$ for $\delta = 10^{-3}$ and 10^{-4} in a numerical check.

9.4 Other results of this phase

- **Locality**: when the law is a non-local operator it is invisible to $\mathrm{Ann}(u)$; an irrational fractional power or a delay gives $\mathrm{Ann} = 0$ (a delay is in the local class through an auxiliary transport) (ledger CONV-T6, CONV-P6, CONV-P7).
- **Dimension** $1 + d$: the ambiguity of the factor decomposition is infinite unless the factors are pairwise co-maximal (ledger CONV-D1); on a convex region $\ker(Q_1Q_2) = \ker Q_1 + \ker Q_2$ in C^∞ (ledger CONV-D2).

- **Boundary conditions** are invisible to $\mathrm{Ann}(u)$; stability holds for a dictionary of traces (ledger CONV-B1–B3). Splitting with a Dirichlet boundary has global order 1 (Lie) and $\Theta(h^{5/4})$ (Strang) in L^2 (ledger CONV-T8).
- **Mori–Zwanzig** for constant-coefficient 2×2 systems: exact memory form and eliminated law (ledger CONV-MZ1–3); the criterion "$E_{\mathrm{int}}=0 \iff [\mathcal{P},L]=0$" is false (ledger CONV-MZ3-F).
- **A convergence contract** for chains of steps: classes (A) exact, (B) convergent with explicit error, (C) no guarantee, (Lim) in the limit; the error of a chain of Lipschitz steps is $\le \sum_i (\prod_{j>i} L_j)\varepsilon_i$ (ledger CONV-T7).

Remark 9.8 (Why it matters). Learning an operator from growing data is possible only in the limit and without a certificate; deciding whether any law exists is impossible even in the limit; and noise can be handled only relative to a finite level of the dictionary and a margin that the data cannot supply.

10 The analytic encoder

Starting from: a Schrödinger-type equation with a variable potential $q(x)$ *known*, written $P(\partial_t, L_q)u=0$ with $L_q=-\partial_x^2+q$ and $L_0=-\partial_x^2$, on a symmetric interval J. *Why:* the variable potential is a nuisance: the theory of Phases 1–5 applies to L_0. Can one change the unknown by an explicit, invertible, stable map so that the potential disappears, without machine learning? Such a map is an *encoder*. *Role:* supplies Step 5 of the extended algorithm (Section 13): the encoder is a *translation* to constant coefficients, not a source of information. The proofs are Dim. A44–A47 and B79.

10.1 What an encoder is

A *structured encoder* is a linear map W acting in x only, with (E1) finite-dimensional kernel on its own domain, (E2′) $WL_q=(L_0+c)W$ for a constant c, and (E3) explicit. The conditions are all needed: without (E1) the zero map qualifies, and without (E3) the interaction-picture map $U_0(t)U_V(t)^{-1}$ does.

Proposition 10.1 (A constant potential has no strict differential intertwiner (ledger ENC-K0) [proved])**.** *For $q\equiv c\neq 0$ no differential operator of any order satisfies $WL_c=L_0W$. With the relaxed requirement $WL_c=(L_0+c)W$ the identity works.* [proof: Dim. ENC-K0]

10.2 The Volterra transformation operator

Let K solve the Goursat problem $K_{xx}-K_{ss}=q(x)K$ on the triangle $|s|\le|x|\le a$, with $K(x,x)=\frac{1}{2}\int_0^x q$ and $K(x,-x)=0$, and put

$$Tf(x)=f(x)+\int_{-x}^{x}K(x,s)f(s)\,\mathrm{d}s,\qquad W=T^{-1}.$$

Theorem 10.2 (The symmetric Volterra encoder (ledger ENC-1, ENC-U1) [proved])**.** *For $q\in C^1$ the problem has a unique solution, and for every $f\in C^2(J)$, without any boundary condition, $L_qTf=TL_0f$ with $(Tf)(0)=f(0)$, $(Tf)'(0)=f'(0)$. T is bijective on C, C^2 and (for $q\in C^\infty$) C^∞, with $\|K\|\le\frac{aQ}{2}e^{a\sqrt{Q}}$ and a Lipschitz dependence on q. Among all Volterra intertwiners the one preserving Cauchy data at the base point is unique; it depends on the base point already for constant q.* [proof: Dim. ENC-1]

Remark 10.3 (Intuition). In characteristic coordinates the kernel solves a Volterra integral equation by successive approximations, with the bound $(Q|\xi\eta|/4)^n/(n!)^2$. Volterra operators with a triangular kernel have no spectrum, so T is invertible, and the Goursat conditions are exactly the ones that make every boundary term in the Leibniz expansion of $L_qTf - TL_0f$ cancel.

Corollary 10.4. *One encoder for every equation* (ledger ENC-C1) [proved] *For $q \in C^\infty$ and every polynomial P: $P(\partial_t, L_q)u = 0 \iff P(\partial_t, L_0)(Wu) = 0$. The same W serves Schrödinger, heat, wave and Klein–Gordon.* [proof: Dim. ENC-C1]

Example 10.5. Constant potential For $q \equiv c$ the transformation maps the exponential $e^{ik\cdot}$ to $e^{ikx_0}[\cos\mu(x-x_0) + ik\sin\mu(x-x_0)/\mu]$ with $\mu^2 = k^2 - c$: the eigenfunction of L_c with the Cauchy data of e^{ikx} at x_0.

10.3 What no encoder can do

Theorem 10.6 (No time-independent encoder for a time-dependent potential (ledger ENC-P0) [proof written, not audited])**.** *Let W be linear with finite kernel and $W(-\partial_x^2 + V(t,\cdot))f = (-\partial_x^2 + c(t))Wf$ for all t. Then $V(t,x) - V(s,x) = c(t) - c(s)$: the potential must be autonomous up to a function of time. The reason is that otherwise infinitely many independent functions $m\varphi_k$ would lie in $\ker W$. Encoders for a genuinely time-dependent potential exist but depend on time (interaction pictures), and a pointwise representative with the identity time change exists exactly for $V = \beta(t)x + \gamma(t)$.* [proof: Dim. ENC-P0b]

Theorem 10.7 (Blind encoders and identifiability (ledger ENC-N1′) [proved])**.** *A blind encoder, one whose output does not depend on which potential produced the data, exists iff the coherence condition (C) holds for the chosen family; it is implied by identifiability and is strictly weaker (a witness: the data model that observes only a phase). The correct statement is: blind encoder $\iff$ identifiability modulo the gauge group of the data model.* [proof: Dim. ENC-N1p]

Remark 10.8 (Warning). [title=An earlier claim that turned out false] The synthesis of the first version, "the exceptions change the time", was false as written: Avron–Herbst and the whole class $V = \beta(t)x + \gamma(t)$ have a pointwise representative with the identity time change. Likewise "a blind encoder exists iff $W_V^{-1}v$ is injective" is false (ENC-N1).

10.4 Other results of this phase

- **Darboux encoders**: a differential W with $WL_q = (L_0 + c)W$ is a Darboux chain on each component of a complement of a countable closed set; for rational $V \to 0$ it exists iff V is an iterated Darboux transform of 0 (Adler–Moser) (ledger ENC-K1′, ENC-K1). They are unbounded with a non-trivial kernel, so they lose modes.
- **The interaction term with a potential**: $\|E^{\mathrm{mult}}f\| \le \frac{t^2}{2}(\|V''\|\|f\| + 2\|V'\|\|f'\|) + \frac{2t^3}{3}\|V'\|^2\|f\|$ with the optimal constant $2/3$, and $E^{\mathrm{mult}} \equiv 0$ iff V is constant (ledger ENC-A9).
- **Faddeev decomposition** on an interval and **Foldy–Wouthuysen** for the free Dirac $1+1$ are decompositions, not encoders (ledger FAD-1, FW-1).

Remark 10.9 (Why it matters). Removing a variable potential is possible by an explicit stable Volterra transformation when the potential is known, and the same map works for every

equation in (∂_t, L_q). When the potential is unknown the encoder gives no information, and for a time-dependent potential no time-independent encoder exists.

11 The universal dictionary and the misfit certificate

Starting from: a filtered dictionary $\mathcal{D}_0 \subset \mathcal{D}_1 \subset \cdots$ of finite sets of operators (the *atoms*), enumerated by an explicit algorithm; a datum u *fits* a level if some non-zero combination of its atoms annihilates u. *Why:* the project was asked: can a dictionary be built that "takes everything into account"? The honest answer has three parts: in the sense of approximation, yes; as exact identification of every operator, no; and in between, a misfit can always be certified by finite data, while a fit never can. *Role:* supplies Steps 2 and 4 of the extended algorithm (Section 13). The proofs are Dim. A48–A51.

11.1 What "universal" can mean

Theorem 11.1 (Every local linear operator is a limit of dictionary elements (ledger DICT-U1) [proved])**.** *By Peetre's theorem a linear operator on $C^\infty(\Omega)$ that does not enlarge supports is a differential operator with smooth coefficients, of locally finite order. With a partition of unity and polynomial approximation of the coefficients it is a limit, in the topology of locally bounded order, of elements of the explicit dictionary $\{\chi_j(y) y^\beta \partial^\alpha\}$. It is a pointwise limit of polynomial-coefficient operators iff its order is globally finite.* [proof: Dim. DICT-U1]

The same holds for classes beyond differential operators: regular non-linear operators (differential polynomials, by a local version of a Peetre–Slovák argument), symbols with loss ε, completely monotone memory kernels, delays (as an augmented transport system) and systems, each with its own explicit atoms (ledger DICT-U2, U3, RIT1, U4).

11.2 What it cannot mean

Theorem 11.2 (No countable dictionary identifies everything (ledger DICT-N1) [proved])**.** *(i) For any countable dictionary the exactly representable operators form a meagre set (Baire). (ii) No finite data certifies the fit or the identification. (iii) If $u \in \mathrm{EP}$, at every high level $\mathrm{Ann}(u) \cap R_{\le n} \supsetneq (L) \cap R_{\le n}$ for any single law L: the kernel is strictly larger than the consequences of one law. (iv) A procedure with finite outputs emits only countably many operators. (v) For a filtered family of differential polynomials the set of u that satisfy some equation of the dictionary is meagre* and *shy (prevalence): the generic u satisfies none.* [proof: Dim. DICT-N1]

Remark 11.3 (Intuition). A compact set in an infinite-dimensional space has empty interior, so a countable family of finite-parameter families covers a "thin" set. For the measure statement one moves u along a direction on which the determinant of the dictionary applied to $u + sw$ is a non-zero polynomial in the single real parameter s: only finitely many s lie in the bad set.

Remark 11.4 (Warning). [title=What this theorem does not say] "Everything" holds in the sense of approximation (Theorem 11.1) plus a misfit certificate, not as exact identification. The first form of the statement for coefficients that are C^∞ functions ("no operator is identified") is false and was replaced by a rank theorem: an affine family is identified iff $\ker \Phi_u = 0$ (ledger DICT-RANGO).

11.3 Fit and misfit

Theorem 11.5 (Asymmetry of fit and misfit (ledger DICT-A, MF1–MF3) [proved])**.** *Let* $\{D_0, \dots, D_N\}$ *be independent continuous operators. The set of* u *with a* misfit *(the* $D_i u$ *independent) is open, and each such* u *has a certificate: a non-vanishing determinant* $\det\langle D_i u, \psi_j\rangle$ *for finitely many test functions, valid in a neighbourhood. The set of* u *that* fit *is closed with empty interior, and for differential atoms a fit cannot be certified even from finitely many* exact *measurements. [proof: Dim. DICT-A]*

Example 11.6. A certified misfit, an uncertifiable fit For $u = e^{-t}\sin x$ the level $\{\partial_t, \partial_x\}$ shows a misfit: the Gram matrix of (u_t, u_x) is non-singular and $\varphi = \langle\cdot, \mathrm{res}\rangle/\|\mathrm{res}\|$ certifies that no first-order law holds. The level $\{\partial_t, \partial_x^2\}$ fits ($u_t = u_{xx} = -u$), and no finite data can certify that.

Misfit certificates exist in three forms: a Hahn–Banach functional (distance to the span), a Gram quotient in a Hilbert space, and a weak form that needs no derivative of u (only integrals against test functions) (ledger MF1, MF2, MF3).

11.4 Stopping rules and the limit

Theorem 11.7 (Stopping rule and identification in the limit (ledger DICT-GR, I-LIM) [proved])**.** *If* n_0 *is the least level with an affine fit and the kernel at every higher level equals the consequences of that law, then the test stops exactly at* n_0*. With a compact class of coefficients and uniqueness, small residual means close coefficients:* $\|L_n - L\| \le \omega(C_u \varepsilon_n)$*, and with data of error* δ *one gets* $\omega(C_u\varepsilon_n + 2C_{R'}\delta)$*, a Hölder rate in the one-coefficient case. [proof: Dim. DICT-GR]*

11.5 Other results of this phase

- **Exact gain of an atom** (ledger DICT-R3); **change of frame** by the Volterra encoder, with exact intertwining and constants $\|W\|, \|W^{-1}\| \le 1 + a^2 Q\, e^{a\sqrt{Q}}$ (ledger DICT-R6).
- **Exponential families**: exact polynomialisation, kinks and the 2-kink (ledger DICT-NS).
- **Minimax rate** $\Theta(\varepsilon^{1/3})$ for the potential reconstruction under $M > \delta_0$ (ledger DM-δ-MINIMAX).

Remark 11.8 (Why it matters). A dictionary can be universal for approximation, never for exact identification. Its negative side is both topological and measure-theoretic; its positive side is a one-way certificate: misfit is witnessed by finitely many integrals, fit by none.

12 What Phase 10 changed

Starting from: the open problems and conditional statements of all earlier phases, reviewed in Phase 10 by three research sessions (phases 1–4, phase 5, phase 8), each followed by an independent audit and, where needed, a second round. *Why:* an honest edition has to say which earlier statements were refuted, which were closed, and which remain open. Phase 10 refuted five of them and closed many. *Role:* in case of a difference with Sections 4, 3.5 and 6 this section prevails. The proofs of the two corrections that change statements are Dim. A52 and A53; the other items are list cards (Part IV).

12.1 Statements that turned out false

Proposition 12.1 (The interaction no-go with constant terms (ledger S10A-T1, P1-Q6) [proved])**.** *Section 4 left open (P1-Q6) whether $u_0 \mapsto E_{\mathrm{int}}$ fails to be continuous in H^m for every operator with at least two non-proportional terms. The statement as written is* ***false****: for reaction–diffusion $u_t = Du_{xx} + ru$ (two non-proportional terms) E_{int} is continuous from H^1, with*

$$E_{\mathrm{int}}(u) = (e^{sr} - 1)\big(v(t) - v_0\big), \qquad s = t - t_0.$$

The correct form is: for monomial terms of distinct degrees, E_{int} is unbounded $H^m \to L^2$ for every m and base time iff there are at least two terms of degree ≥ 1*; with at most one the interaction is bounded for every $m \geq 1$. Constant terms only multiply the solution by a bounded exponential and neither create nor destroy the no-go. [proof: Dim. S10A-T1]*

Proposition 12.2 (Holes: the sum of kernels is closed but not dense (ledger S10A-P1) [proved])**.** *On the annulus $A = \{1 < |z| < 2\}$ the Laplacian is a product of the two coprime factors $\partial_t \pm i\partial_x$. The harmonic function $\log|z|$ is not in the closure of the sum of holomorphic and anti-holomorphic functions in any H^m, and for $m \geq 1$*

$$\ker{}_{H^m(A)}\,\Delta = (\mathcal{H}_1 + \mathcal{H}_2) \oplus \mathbb{C}\log|z| :$$

the sum is closed, of codimension $b_1 = 1$, and not dense. Hence the density question of Section 3.5 has a negative answer on non-convex regions in general (P2-Q1 is false as a universal statement), and this is a differential *example of "closed but not dense". [proof: Dim. S10A-P1]*

Remark 12.3 (Intuition). A hole gives a period: the flux of a harmonic function through a loop around the hole. The flux is a continuous functional that vanishes on holomorphic and anti-holomorphic pieces (their derivative has a single-valued primitive), but not on $\log|z|$.

Three more statements were refuted: the extension of the convex density theorem to non-convex regions (S10B-C1, by a hole with an interior); P2-Q1 as a universal statement (for Δ on every Lipschitz region with $b_1 \geq 1$); and, in the variable-coefficient phase, "$P11(e)$ with B merely continuous" and "the pole depends only on $A(\xi_0)$ and the factorisation".

12.2 Problems closed

- **Density of the sum of kernels** on every simply connected Lipschitz region, and for $\delta(P) = 0$ on every Lipschitz region, modulo two cited facts [F1], [F2]; the source of the collar field was replaced by a proof (ledger S10A-T3, S10COL).
- **C12-P7**: "closed but not dense" realised by the annulus.
- **C3-Q1**, the decidability question, closed as a *classification by data model* (composition of black-box impossibility, bound, analytic and jet models); no new mathematics.
- **OPF5C-T5, INT-T1(B), S10B-T2**: the multi-soliton is a Hirota solution; the reflectionless Jost computation holds.
- **Uniform shock rate with several births** (OPA-Q1): modulo the equivalence Oleinik $\Leftrightarrow$ Kružkov [F1].
- **Variable coefficients**: VAR-P11(e) by Volterra and Gronwall; a solution with $V = 0$, $|Z| > 0$ and $\operatorname{int} Z = \emptyset$ (Tychonoff and a Cantor-type set); Floquet in the quadratic class ($\operatorname{tr} M = -26/5$); a monic criterion in the Weyl algebra.

- **Systems**: Maxwell has an identifying field; at fixed real time no finite product of Dirac flows with polynomial times; the pole criterion $E_0 \cap F_0 \neq 0$.
- **Dictionary**: the L^1 rate for general completely monotone memory kernels; the minimax rate $\Theta(\varepsilon^{1/3})$.
- **Computer-assisted**: $N \leq 4$, and, after the triage, $N = 5$ and $\delta_6(12) = 7$ for the syzygy degrees (modulo a novelty hypothesis (Nov)); $\delta_6(11) = 8$ with a written proof.

12.3 Textbook examples and formalisation

Five textbook examples were checked against the whole theory: free Dirac $1+1$, the harmonic oscillator, the Pöschl–Teller potential, the fractional heat equation and Maxwell in vacuum; 80 of 80 checks pass, and three new arguments were proved. The formalisation in Lean 4 with Mathlib reached 37 entries: 27 are faithful to the ledger statement, 7 are weaker and 3 are different, none is vacuous, and all 85 `#print axioms` use only the standard axioms. Mathlib has no Weyl algebra and no C_0-semigroups, so the analytic parts are not formalised.

Remark 12.4 (Why it matters). The corrections of Phase 10 all push in one direction: the negative results of Phases 1–3 are sharper than first stated (constants do not break the no-go; holes do break density), and the closures are explicit about what they depend on.

13 The extended algorithm

Starting from: the results of Sections 7–11 and of the algorithm of the companion paper; a datum in one of the models below. *Why:* the algorithm of the companion paper turned the theorems of Phases 1–5 into a seven-step procedure for a scalar constant-coefficient equation. This section extends it to systems, variable coefficients, noise, and operators outside the catalogue, and states, for every step, what is promised and what is not. *Role:* the operational summary of the whole edition. It introduces *no new claim*: it composes rows of the ledger.

13.1 Which senses of "identify"

The word "identify" has five senses, and every step declares which one it uses.

Sense	Meaning	Unique when
ID-ann	$\mathrm{Ann}(u) = (P)$: the annihilator ideal is principal and generated by the law	always if proved; never for $u \in \mathrm{EP}$ (the point spectrum is non-empty)
ID-ker	$\ker \Phi_u = 0$ in a fixed parametric class	only if a law of the class exists (H-ex)
ID-res	coefficients determined by the residue map, e.g. $V = (iu_t + u_{xx})/u$	on $\{u \neq 0\}$, in the classes of Section 8
ID-fam	identification modulo a family: $V + c$, a gauge class, an encoder modulo $c(t)$	–
ID-lim	identification in the limit, never certified at finite time	under (H-ID)

13.2 Data models and the contract

Model	Datum	Senses allowed
DM1	a closed form that is an exponential-polynomial	only ID-ker relative to a finite class, never ID-ann
M3-hol	a closed-form holonomic non-EP function (Bessel, Airy, sech, Gaussians)	ID-ann, ID-res, ID-ker, with a symbolic proof
DM2	the jet at a point	only ID-lim: no certificate at finite time
$M2^+$	jet plus a zero test, or a declared holonomic ideal	exact decidability modulo (AC)
DM3	u known exactly in $H^m(\Omega)$ or $C^\infty(\Omega)$	ID-ann, ID-res, ID-ker, ID-fam
DM-δ	$\tilde{u}$ with $\|\tilde{u} - u\|_Y \le \delta$, norm Y declared	no identification: only the moduli and conditional stability

The contract has four classes: (A) finite and exact, (B) convergent with an explicit error ε_i, (C) no guarantee, (Lim) in the limit without a certificate. The pipeline has **two layers**.

1. *Conditioning* (Steps 1–3) fixes the model, the level and the class. Its outputs are discrete, and since $\mathrm{Ann}(\cdot)$ is discontinuous everywhere (Theorem 9.6) no Lipschitz bound applies to them. Completeness ("the right law is in the chosen level") is always (C).
2. *Conditional correctness* (Steps 4–6) has metric outputs. *At the declared level and in the declared class* the error of the chain is bounded by $\sum_i (\prod_{j>i} L_j)\varepsilon_i$, provided every step is (A) or (B) and Lipschitz in its computed inputs.

13.3 The steps

Step	Task	Result relied upon	Output
1	model	Thms. 5.5, 5.5, 9.1; Prop. 7.14; Thm. 8.9	model, ring of coefficients, senses reachable
2	level of $\mathcal{D}_\infty$ (L0–L10)	Thm. 11.2	level n and finite dictionary, or `NON-DECISO`
3	identification (scalar, system, variable)	Chs. 5, 7, 8	$\mathrm{Ann}(u) \cap R_{\le n}$, G_u, σ_p, sense
4	misfit certificate or best approximation	Thms. 11.5, 11.7, 9.6	`MISFIT`, `FIT-COMPATIBLE` or stop
5	analytic encoder for V	Thm. 10.2, Cor. 10.4	translated datum and certificates, constant C_W
6	factors, terms and E_{int}	Chs. 3–4, 12	components, ambiguity $\delta(P)$, E_{int}
7	convergence guarantees	Thm. 9.1–9.6	double label and chain bound

Remark 13.1 (The levels L0–L10). L0–L2 scalar constant coefficients; L3 non-monic or degree ≥ 2 in τ; L4 polynomial non-linear; L5 memory and delay (augmented system); L6 constant-coefficient systems (Dirac, Lamé, Maxwell); L7 variable coefficients in x (Schrödinger, Airy); L8 coefficients in (t, x) (Stark, oscillator, Floquet); L9 non-local (fractional, multipliers); L10

Table 4: The kinds of claim made in the paper.

Kind	Statements
proved	groups are factors with overlap $\delta(P)$ (Theorem 3.1); terms are not groups (Propositions 3.3 and 4.2); identification criterion and its limits on finite data (Theorem 5.1 and Propositions 5.2 and 5.5); stability via γ_N (Theorem 3.4); Bayesian error bars via the angle (the companion algorithm paper); counting condition
design rule	labels instead of "identified"; distributions of ρ instead of values; decisions on the raw observations; growth of the class instead of coefficient penalties; minimum-norm split of the overlap
measured	0/20 confident wrong answers on random equations not used in development; 60/60 factor sets on grid designs with independent fields and noise; 30/30 on the development benchmark; coefficient coverage 0.94 at nominal 0.90

general dictionary with misfit and stopping. The choice of level is a *heuristic* (completeness (C)): Theorem 9.3 excludes any a priori criterion.

Remark 13.2 (Step 3 by case). **Scalar** (L0–L5): the tests of Section 5; in DM1 never ID-ann. **Systems** (L6): the existence test (socles ≤ 1, Theorem 7.2), then, for a given u, ID-ann or, in Hamiltonian form, ID-ker with a pointwise linear system of rank 8 (Dirac $1+1$) provided at least four generic modes. **Variable** (L7–L8): the residue map ρ_u (two derivatives and a division per point) or the affine set $\theta_0 + \ker \Phi_u$.

Remark 13.3 (What the pipeline never outputs). It *never* emits "law identified" from finite numerical data. Its outputs are: `MISFIT(certificate)` at level n; `FIT-COMPATIBLE(`n, r_n, `bound)`; `IDENTIFIED[sense, model]`, only with an exact symbolic proof in DM3, M3-hol or M2$^+$ (modulo (AC)), or relative to a finite class in DM1; `ID-LIM(`n`)`, a current candidate without certificate; and `NON-DECISO`, when the budget runs out.

Remark 13.4 (Warning). [title=The encoder is a translation] Step 5 is only usable when q is *given*. If q is estimated at Step 3c the Lipschitz constant of $q \mapsto W_q$ is not explicit and the transport is (C). The only proved bound is $C_W = 1 + a^2 Q\, e^{a\sqrt{Q}}$, which can overestimate by about four orders of magnitude at $a\sqrt{Q} = 5$; a sharper bound (a cosh or an exact Gronwall estimate) is open. A Darboux encoder is unbounded with a non-trivial kernel: it loses modes, and an identifying datum can become exponential-polynomial.

Remark 13.5 (Why it matters). The extended algorithm is two things at once: a procedure and a map of its own limits. What it can promise is correctness *at a declared level and class*; what it can never promise is that the level is the right one. That asymmetry, not a technical gap, is the content of the negative theorems of Phases 1–8.

14 Limits, open problems, and status

Starting from: no new assumptions. *Why:* to state what is impossible, what is open, and how reliable the results are. *Role:* delimits the claims of the paper.

What is impossible (all [proved]).

- finite certificates of identification, and black-box identification (Section 5.2);
- per-term decomposition of the solution (Proposition 3.3);

- stable regularization of E_{int} (Proposition 4.2);
- identification by restricting variable-coefficient operators to one solution;
- decomposition on domains with holes (Figure 4);
- a finite product of single-term flows reproducing non-commuting dynamics.

Open or conditional.

- identification for analytic KdV data with nonzero reflection coefficient (ledger INT-Q1) [open];
- time-uniform shock estimate with several shock births (ledger OPA-Q1) [conditional];
- density of the sum of kernels on nonconvex domains [open];
- exact values $\delta_N(K)$ of an invariant for general N (ledger OPF5C-Q1) [open];
- nonlinear asymptotics with derivatives (ledger S4ID-P8) [conditional];
- closedness of the sum of kernels for $N \geq 3$ differential factors (ledger C12-Q2) [open].

Honest assessment. Many ingredients are classical (Malgrange–Ehrenpreis, Nullstellensatz, Cole–Hopf, inverse scattering). The contribution is the *package*: sharp characterizations, negative theorems, and counterexamples, each with a verified status. Independent reviewers were AI agents. Fifteen theorems of the algebraic core (the E_{int} identity for commuting flows, the comaximal factor decomposition with explicit projector, companion normality for $k = 2$, irreducibility results) are formalized in Lean 4/Mathlib without `sorry`; the analytic parts (distributions, Sobolev spaces, semigroups) are not, since Mathlib lacks the libraries. Several citations were verified only on secondary sources; a result relied upon should be traced to its primary source. Out of scope so far: a variable potential treated as noise (Schrödinger/Dirac), matrix-valued coefficients (4×4 Dirac), and any statistical preprocessing.

14.1 Limits of all eleven phases

Starting from: no new assumptions. *Why:* to state what is impossible, which statements turned out false, what remains open, and how reliable each result is, for the whole project (Phases 0–10). *Role:* delimits every claim of this edition. The analytic part of the project is *closed* by decision of the user (28 October 2026); the unresolved items are collected in the project file `PROBLEMI_APERTI.md`, none of them bearing a central result.

14.1.1 What is impossible

Each item below is a proved negative theorem, not a gap.

- a finite certificate of identification, and identification from a black box (Theorems 5.5 and 5.5); in the limit, no learner decides whether a law exists (Theorem 9.3);
- identification by restricting variable-coefficient operators to one solution (the restriction result of the complete edition); a huge class of coefficients identifies nothing (Theorem 8.1);
- decomposition of a solution by additive terms (Proposition 3.3); factor decomposition on regions with holes (Proposition 3.5);
- components in H^m in general: one derivative can be lost (Counterexample 3.6); density of the sum of kernels on regions with holes (Proposition 12.2);
- a continuous or stably regularised interaction term in H^m *when at least two terms have degree* ≥ 1 (Theorems 4.2, 4.2, 12.1);
- a finite product of single-term flows reproducing non-commuting dynamics when the adjoint orbit is infinite (Theorem 4.10(c)); a uniform finite product for Dirac $1 + 1$ with transport and mass times independent of the frequency (Proposition 7.13);
- an identifying solution when a module has a socle of dimension ≥ 2 (Theorem 7.2); one field identifying (A, V) in the magnetic Schrödinger equation (Proposition 7.11);
- a time-independent encoder for a genuinely time-dependent potential (Theorem 10.6);
- exact identification of every operator by a countable dictionary (Theorem 11.2), and any finite certificate of a fit (Theorem 11.5).

14.1.2 Statements that turned out false

Statement	Counterexample or correction	Ledger
the spectrum of an expansion decides identification on a bounded domain	e^{x-t} on the unit square	C3-C1
components keep the regularity of u	parallel transports on the square	C12-C1
$u \mapsto E_{\mathrm{int}}$ is continuous in H^m	real exponentials in advection–diffusion	P1-P1(e)
the companion matrix is normal iff $p = s^m - c$	$s^2 + s - 1$	R6-T3
splitting by terms lies in $\ker(L_1, L_2)$	$\frac{1}{2}x^2 + Dt$ for the heat equation	R2-T3 (corrected)
exact interaction impossible for infinite Lie algebras	$e^{-tT_2}e^{-tT_1}e^{t(T_1+T_2)}$	R5-Q5, P1-C1
NLS identification by scattering data without multiplicity	explicit counterexample	OPF5D
no continuity for *any* operator with two non-proportional terms	reaction–diffusion $u_t = Du_{xx} + ru$ (Theorem 12.1)	P1-Q6
density of the sum of kernels on every Lipschitz region	the Laplacian on the annulus (Proposition 12.2)	P2-Q1, S10B-C1
$\Phi(V, v) = W_V^{-1}v$ injective $\iff$ a blind encoder exists	the phase-only data model	ENC-N1
an eigenvalue crossing implies a pole of the projector (derogatory A)	$A = \mathrm{diag}(\xi, -\xi)$ has constant projectors	SYS-II-T2-C
every $u \not\equiv 0$ identifies $V(t, x)$	Tychonoff silent strips	VAR-T1(c)-F
the chirp–free–chirp word breaks at the lens time	explicit continuation	VAR-T10-F
Dyson–Phillips with B merely continuous	a rank-2 counterexample	VAR-Q18-CE

14.1.3 Open or conditional

These are not closed and are not declared closed. For each the ledger records what is proved.

With a proved reduction or condition.

- the Lie-flow question for algebras with (H) but without a weighted dilation [open] (ledger P1-Q7) (candidate $(x^3 + x)\partial, \partial$ proved);
- the closedness of the sum of kernels for $N \geq 3$ differential factors, reduced to triples with at least two non-compact pairs, outside real transports [open] (ledger C12-Q2);
- the graded case of the syzygy degrees [open] (ledger OPF5C-Q1); the realisability of atoms [open] (ledger OPR2-Q3); the sufficient condition (L-v) at every vertex [open] (ledger OPR2-Q2′);
- identification for analytic KdV data with nonzero reflection, equivalent to (R), modulo [GT-u$_m$] and (WP) [open] (ledger INT-Q1);
- boundedness of the Hölder constants in the Gelfand–Levitan analysis [open] (ledger [GT-u$_0$]); the justification of a differentiation in a cited analysis [open] (ledger [GT-u$_m$]).

Without a known route. Non-Lipschitz domains (P2-Q1-NL: only a model case is proved); the exact codimension for a general P (P2-Q1-COD); closedness of the sum for $(\tau - \xi^2)(\tau + c\xi)$ (C12-Q1); multiplications and a general Floquet logarithm (VAR-Q18, VAR-

P12(d)); the non-monic case in the Weyl algebra (VAR-C3Q2-open, VAR-Q13); complex-time Dirac with polynomial times (SYS-DIRAC-POL); the DM3 reading of ENC-Q4 and the real residue ENC-Q10; the optimality of the L^1 rate for general kernels (DICT-U3d-v); the C^∞ division reformulation of DICT-Q3(c).

Conditional on sources not read in full. Oberst 1993 (read only as an abstract; it concerns novelty, not a load-bearing step); the equivalence Oleinik ⇔ Kružkov [F1] (a plan exists: vanishing viscosity on Cole–Hopf, no audited proof); Malgrange 1966; the existence of global solutions in $\mathcal{S}$ (WP); Björk. The project forbids downloading files, so these items are *blocked on sources unavailable offline*.

14.1.4 How reliable the results are

Proofs and audits. Every statement labelled [proved] has a complete proof that passed an independent review. The reviewers were AI agents with no access to the author's reasoning, not human referees. Statements labelled [proof written, not audited] have complete proofs that were not audited. For Phases 6–10 the labels follow the ledger: a result is [proved] if the ledger records it as proved after the audits of Gate 8 and of Phase 10 (S-CHECK-F8, S-CHECK-F10). **The proofs in this edition, however, were rewritten from the research packages for this volume (cards A32–A53 and B73–B79) and were not themselves re-audited**; the label refers to the result, not to the rewriting. Several packages cite theorems (Peetre, Malgrange, Oberst, Marchenko) whose exact statements were verified only on abstracts or secondary sources.

Formal verification. Fifteen theorems of the algebraic core of Phases 0–5 and, in Phase 10, 37 entries (27 faithful, 7 weaker, 3 different, none vacuous, 85 `#print axioms` with standard axioms only) are formalised in Lean 4 with Mathlib. The analytic parts are not.

Citations. Many ingredients are classical. The primary sources of the load-bearing citations of Phases 0–5 were checked; for Phases 6–10 a number of citations were checked only on abstracts or secondary sources (Oberst, the transformation-operator theorem numbers, Casas for Magnus). A result relied upon should be traced to its primary source.

Novelty. The contribution is the package: sharp characterisations, negative theorems and counterexamples, each with a verified status. Parts of the identification theory are folklore in commutative algebra (Oberst duality, socle criteria, Peetre's theorem, Volterra transformation operators); parts of the finite-data theory are known in substance (Prony, Mourrain).

14.1.5 What this edition does not contain

- The proofs of every result of Phases 6–10: the catalogue contains proofs of 22 results in Part II and of 25 more in Part IV; the other 86 results appear as list cards with their trust label and the research file the companion catalogue of the catalogue onwards).
- The catalogue of about 60 equation types and the tropical reading of scaling limits.
- The Lean sources.
- Phase 11 as a research phase: this edition *is* its deliverable. The text written for it (the rewritten proofs of Parts II and IV, the corrections added to Sections 3.5 and 4, and the reorganisation of the extended algorithm in Section 13) has not had the proof-checker audit, citation audit and kill-argument review that the plan required before closing.

15 The complete list of open problems

Starting from: the registry `PROBLEMI_APERTI.md` of 28 October 2026, the closing report of Phase 10, the final report of Phases 0–5, the open-question files of every research package, and Section 11 of the extended algorithm. *Why:* Section 14.1 states what is open in a few lines. This section is the full registry: every open or conditional item of the project, with what is already proved about it and what would be needed to close it. Nothing here is a central result. *Role:* a starting point for any later research phase. The status of every item in the ledger is unchanged: a label is raised only after a proof and an audit.

15.1 Why the analytic part can be closed

None of the items below carries a central result of the project (the linear characterisations, the algebraic theorems ALG-*, the dictionary criterion S4ID-T1, the sum-of-kernels theorems C12-T3 and P2-T3, the negative theorems and the counterexamples). Each item is of one of three kinds: (i) an extension to more general classes of a result already proved in a model case; (ii) a new question born during the work; (iii) a dependence on a published source that was not available offline.

The only items from which statements of the report depend are listed with their dependence, and each is already marked "modulo" or "conditional" where it is used:

- **[F1] Lax–Oleinik = Kružkov.** The source is classical and was read (Kružkov 1970, Thm 1, §3); what is missing is an autonomous proof, for which the triage gives a plan (vanishing viscosity on Cole–Hopf). Burgers beyond the shock depends on it (S4T-T3(iii), OPA-T1, OPF5A-T1/T2/C1). This is *not* an open problem.
- **[L-loc]**: the localisation lemma on which OPA-Q1 depends for several shock births; the case of a single shock birth (for example $\sin x$) is closed modulo [F1].
- **[GT-u]**, that is [GT-u_0] and [GT-u_m]: S4ID-P8 and OPR2-P7 (asymptotics with derivatives).
- **[F4$^-$]**: INT-T1(B) (reflectionless N-solitons), verified for $N = 2$.
- **Malgrange 1966**: C3-P1 for general data on convex sets; the jet case is proved without it and the cited theorem is verified.

15.2 Research problems with no known route

Item	Question	Already obtained	What is needed
C12-Q2 = OPF-Q2	closedness of the sum of kernels for $N \geq 3$ differential factors	false in abstract Hilbert space (counterexample); $N = 2$ proved	a differential family with ≥ 2 non-compact interactions
C12-Q1	closedness for $(\tau - \xi^2)(\tau + c\xi)$	—	boundary-layer estimates (route OPR2-L1, heat × transport)

Item	Question	Already obtained	What is needed
OPF5C-Q1	syzygy degrees $\delta_N(K)$ for general N	nine exact values, computer-assisted; $\delta_6(12) = 7$ new; conjectures $\delta_N(2N-1) = N+2$, $\delta_N(2N-2) = N+4$, $\delta_N(2N) = N+1$	a structure for general N; for fixed N it is a finite computation; $\delta_6(11) = 8$ has a written proof, Hirota $N = 5$ is not concluded (a run of about 17 minutes was interrupted)
OPR2-Q3	repeated socle, "no finite test"	proved under (I) and (II)	a lifting theory of atoms
OPR2-Q2′	wedges with degenerate directions	tangent model closed and direct (OPF5B-P8); numerical check 7/7	(L-int): an explicit Hilbert–Schmidt estimate, a few hours of work; (L-v): analysis at the vertex of the cone
INT-Q1	KdV, $r \not\equiv 0$, u analytic: is there an autonomous ODE in x?	OPF5C-T6: order ≤ 3 implies 0 or a 1-soliton	singular relations at the origin and sub-dominant terms of the radiative asymptotics (upstream: [GT-u$_m$])
P1-Q7, general	Lie algebras with (H) but without a weighted dilation (Dil)	the candidate $(x^3 + x)\partial, \partial$ is proved	no route without a dilation
P2-Q1-NL	non-Lipschitz domains	Δ, $m \geq 1$, $L^{1,2} \subset L^2$ proved	general P and $m = 0$
P2-Q1-COD	exact codimension for general P	—	continuous period functionals in H^m
VAR-Q18	multiplications	a rank-2 counterexample with B merely continuous	a counterexample with rough W, v merely continuous, and control of the whole Dyson series
VAR-P12(d)	non-quadratic Floquet logarithms	partial observations (parity, dilations)	an obstruction for non-quadratic logarithms
VAR-C3Q2-open, VAR-Q13	non-monic case in the Weyl algebra	monic case (a′) proved (VAR-C3Q2-R)	a decomposition without monicity
SYS-DIRAC-POL	polynomiality for complex t, m	real t, m proved	the triage sketches a lemma E′ "by growth"; not audited
ENC-Q4	the DM3 reading	—	the class $W(t)$ must be defined first
DICT-Q3(c)	C^∞ division at variable rank	—	a general division criterion
AM-CL	closure AM	no longer needed (ledger F10-TRIAGE)	—
OPA-Q2	density of the Morse+(SB) class	a sketch (Sard with finitely many parameters)	write and audit the sketch

Item	Question	Already obtained	What is needed
non-convex regions	density of OPF5B-T1 outside convex sets	convex case proved; the extension is false on regions with holes (A53); simply connected Lipschitz modulo F1, F2	—

15.3 Items blocked on sources unavailable offline

The project forbids downloading files. Each item below is unblocked by placing the indicated text in the project folder.

Item	Source needed	After reading
[GT-u_0]	Grunert–Teschl §5, Krüger–Teschl	rewriting with explicit constants; a few days of work
[GT-u_m]	Deift–Zhou, CMP 165 (1994) 175–191; Buslaev–Sukhanov 1986	term-by-term derivation; not short
VAR-Q17(b)	Baudouin–Puel; Lasiecka–Triggiani–Zhang; Mercado–Osses–Rosier (Carleman for Schrödinger with $V(t,x)$)	an adaptation
ENC-Q10	Matveev–Salle; Bagrov–Samsonov (pairwise Darboux)	an adaptation, plus a decision on the class $\mathfrak{E}$
DICT-U3d-v	Braess; Braess–Hackbusch	for $t^{-\beta}$ probably the τ_k are not needed; the uniform limit on the class stays a hard problem
Malgrange 1966	*Ideals of differentiable functions*	only to have an autonomous proof: the theorem is already cited
(WP)	Bona–Smith; Kato 1983; Lax 1975	alternatively an autonomous proof of several days
[F4^{-}]	the primary source for reflectionless N-solitons	verified for $N = 2$
Oberst 1993	MCSS 6 (read only as an abstract)	*priority and novelty only*: SYS-EX is proved on an arbitrary Ω without it

15.4 Open questions of each research package

Systems (Phase 8).

- **Q-S8SYS-1** (closed for real times; complex times open): finite words of Dirac flows with times polynomial in ξ.
- **Q-S8SYS-3**: for fixed Ω, does (i$\Leftarrow$) of SYS-S1 hold for every pair iff $\mathrm{Ext}^1(N, C^\infty(\Omega)) = 0$ for every N that occurs, i.e. iff Ω is P-convex for all P ($C^\infty(\Omega)$ injective)?
- **Q-S8SYS-4** (derogatory A): the candidate "the projector has a pole at ξ_0 iff $A(\xi_0)$ has a non-trivial Jordan block mixing eigenvalues of coprime factors"; not proved, high cost; the pole criterion $E_0 \cap F_0 \neq 0$ was proved in Phase 10.

- **Q-S8SYS-5** (novelty of SYS-EX): read Oberst 1990 §2–4, Oberst 1995 (*Variations on the fundamental principle*) and Pillai–Shankar 1998 to decide whether SYS-EX on $\mathbb{R}^n$ is already stated.
- **Q-S8SYS-6** (Maxwell): socle of vacuum Maxwell in (E, B), an over-determined 8×6 system; closed in Phase 10 ($\mathrm{pd} \le 2 \Rightarrow \mathrm{soc} = 0$) in the symmetric class; in the general Hamiltonian class ID-ker is impossible (EX10-e-K).
- **Q-S8SYS-7** (SYS-N2): the determinantal reduction for massless Weyl needs the class group of the light cone (Knörrer/Eisenbud); not touched.

Variable coefficients (Phase 8).

- **VAR-Q10** (closed in Phase 10 by an example). **VAR-Q6′**: the Galois group of $-\partial^2+x^2-E$ over $\mathbb{C}(x)$ (SL_2 for $E \notin \pm(2\mathbb{N}+1)$, the Borel group otherwise); not needed for identification; tool: Kovacic or a direct argument.
- **VAR-Q12**: for L monic of order $m \ge 2$ in ∂_t, an intrinsic characterisation of B-independence (holonomy of $A_2 u$, characteristic dimension); is there a non-D-finite u that does not identify? **VAR-Q13**: for non-constant leading coefficient in every direction, when is $\mathrm{Ann}(u)$ equal to the saturation $A_2 \cap \mathbb{C}(t,x)\langle\partial\rangle L$? A decision procedure needs factorisation of D-modules (Grigoriev–Schwarz).
- **VAR-Q11**: for generic periodic $V(t,x)$, does $U(T,0)$ have a differential (unbounded) logarithm? Known for $V = v(t)$ and quadratic cases.
- **VAR-Q8, Q8′**: convergence of the Magnus series for unbounded operators; when $Y(t;\varepsilon)$ is normal for complex ε, as Casas 2007, Thm 3.4 requires in infinite dimension.
- **VAR-Q14**: whether Saut–Scheurer (1987) or Isakov (1993) contain VAR-L1 for $V \in L^\infty(t,x)$ in 1D (probably VAR-L2 and VAR-T1(c)(ii)); if so, cite them. **VAR-Q15**: read Schindelar–Levandovskyy–Zerz (2011) to confirm the delta of the Weyl-algebra comparison.
- **VAR-Q16**: for continuous matrix V, does "V determined on an open U" imply $\mathrm{span}_t \psi(t,x) = \mathbb{C}^2$ on a dense subset of U? **VAR-Q17**: lateral observability for mild solutions ($\psi_0 \in L^2$, $V \in L^\infty$). **VAR-Q18**: see above. **VAR-Q19**: robustness in DM-δ: construct an estimator of V with guaranteed error, or prove that the Hölder exponent $1/3$ of VAR-P3 is also optimal for the estimation problem (settled with $M > \delta_0$ in Phase 10).

Encoder (Phase 8).

- **Q1**: ENC-U1 in wider classes (kernel $K \in L^1$ for $q \in L^1$, non-triangular kernels). **Q2**: boundedness of W on $H^{2k}(J)$ with explicit constants (the induction with $(Wf)'' = W(f'') - W(qf)$ is only sketched; it is needed for rule R6 with Sobolev norms).
- **Q3** (ENC-S1): transport of the factor decomposition for coprime factors in $S = \mathbb{C}[\tau,\sigma]$ through φ and C12-T1′; to be written out, and to check whether $\mathrm{lcm}(Q, \tilde{Q})$ is the generator.
- **Q4**: a natural class of potentials with $\mathcal{S}_V \cap \mathcal{S}_{V'} \neq \{0\}$ for $V \neq V'$ and a structured encoder for each, so that separation (S) is non-empty. **Q14**: the same modulo the gauge of the data model.
- **Q5**: E^{mult} for Schrödinger on a bounded Ω with Dirichlet conditions (leading term $t^2/2\,[H_0, V]$ presumably, to be proved). **Q6**: ENC-1 in dimension $d \ge 2$: no triangular Volterra operator with local kernel (the characteristic problem is not Goursat).
- **Q7**: does the local Darboux chain extend to all of J across the zeros of the seeds, and is Z discrete (it is only shown closed and countable)? Known for rational potentials (Duistermaat–Grünbaum).

- **Q8**: a quantitative bound for the truncated Foldy–Wouthuysen series with $V \in W^{k,\infty}$ (error $O(m^{-k-1})$?). **Q10**: non-local but explicit encoders for non-quadratic V (Bluman–Shtelen classify only pointwise encoders).
- **Q11**: W commuting with ∂_t with non-constant $c(t)$ and W non-local in t (for example convolutions in t). **Q12**: lateral continuation and (E1) on solutions. **Q13**: regularity of the propagator with V merely norm-continuous (Kato asks for $t \mapsto H(t)$ strongly C^1). **Q15**: ENC-T2(c′) in the class of principal annihilators (proved for the specific family in Phase 10).
- **Q9**: theorem numbers to verify (Marchenko ch. 1, van der Put–Singer, Duistermaat–Grünbaum, Casas).

Convergence (Phase 8).

- **CONV-Q1**: without an upper bound, is $G_u = \gcd \mathrm{Ann}(u)$ identifiable in the limit when $\mathrm{Ann}(u)$ is not principal? (the rule converges to g_d, which can be a proper multiple of G_u). **CONV-Q2**: an explicit jet threshold K^* for classes wider than EP (finite growth order). **CONV-Q3**: a quantitative version of T2 in terms of the complexity of changes of mind.
- **CONV-Q4**: characterise $\mathcal{N}_{\mathcal{K}}$ when $\mathcal{K}$ is the class of augmented systems with finitely many auxiliary variables. **CONV-Q5**: the decomposition $\ker(Q_1 Q_2) = \ker Q_1 + \ker Q_2$ on non-convex domains in $1 + d$ (P-convexity, Hartogs–Ehrenpreis phenomenon). **CONV-Q17**: injectivity of $\mathcal{D}'(\Omega)$ on convex Ω (Malgrange calls it probable; to check in Palamodov 1970).
- **CONV-Q7**: T4 and T5 with stochastic noise, rates in probability. **CONV-Q8**: explicit margin ρ_d for concrete families (heat, waves) as a function of Ω. **CONV-Q14**: T4(b) and T5 for non-integer s (complex or real interpolation; withdrawn from the statement).
- **CONV-Q11**: Mori–Zwanzig with variable coefficients ("closed local law $\iff$ finite elimination" proved only in the direction $\Leftarrow$ with constant coefficients); look for a Weyl-algebra / D-module version. **CONV-Q12**: the L^∞ order of Strang in the P3-T3 framework (numerically 1, an example, not proved). **CONV-Q13**: higher orders in ε for CONV-T8 and non-modal data in H^1_0 or H^2. **CONV-Q6′**: the norm of the orders 1 (Dirichlet) and 1.5 (Neumann/Robin) in Dang–Einkemmer–Ostermann 2025. **CONV-Q16**: identification of a boundary condition with positive margin through a quotient dictionary for tangential consequences (CONV-B3).

Dictionary (Phase 8).

- **Q1**: a uniform-in-t rate for general completely monotone kernels (the explicit rate is known for $t^{-\beta}$; closed in Phase 10; optimality is open). **Q2**: Hölder for $s \geq 1$ in I-LIM needs a Gagliardo–Nirenberg interpolation $\|f\|_{C^s} \leq C\|f\|_{L^2}^{\theta}\|f\|_{C^{s,1}}^{1-\theta}$, $\theta = 2/(2s+2+N)$, on Lipschitz domains with a constant (the exponent is verified by scaling; the source is missing).
- **Q3**: the intermediate case of RANGO (the pointwise residue map $A_u(y)$ is neither injective on a dense set nor onto everywhere): is identifiability decided by division at variable rank, or are there counterexamples? **Q4**: Theorem A(iii) for non-local atoms (memory, symbols, fractional): the proof uses locality.
- **Q5**: order optimality of polynomials via Kolmogorov n-widths (Pinkus 1985, source not read). **Q6**: the 2-kink has kernel 1 (only checked on examples; a proof needs exact independence of $u_{xt}, u_{xx}, u_{tt}, u$). **Q8**: prior art for N1(v)-ALG, "the generic u is differentially transcendental". **Q9**: R6 for $Y = H^{2k}$ and transport of the weak form MF3 with compactly supported test functions.

- **Q7**: theorem numbers and citations to verify: Hörmander III 18.1.13; Trefethen–Weideman (trapezoid rule); Boman 1967; Christensen 1972 and Hunt–Sauer–Yorke 1992 (prevalence); Ax 1971; Weiss–Tabor–Carnevale 1983; Whitney extension.

15.5 Open questions of the extended algorithm

The twelve questions of Section 11 of the extended algorithm:
1. robustness in DM-$\delta(Y)$ (VAR-Q19);
2. an example at level L9 with infinitely many modes and an identified non-local atom, and DICT-A(iii) for non-local atoms;
3. DICT-ILIM-Hs (closed in Phase 10 with an elementary cone lemma);
4. CONV-Q1: G_u in the limit without an upper bound;
5. VAR-T1(c)-Q10 (closed by an example in Phase 10);
6. C3-Q2 in general;
7. CONV-Q9: unbounded V;
8. the cost of rule R1 per level; the audit of C3-A1 (elimination at polynomial cost, a written proof without a ledger row); a rigorous inclusion for the misfit (A);
9. rule R6 in H^{2k} and with the weak form MF3; a bound on $\|W_q\|$ tighter than C_W (at $a\sqrt{Q} = 5$ the proved $C_W^2 \approx 1.4 \cdot 10^7$ against $\cosh^2(a\sqrt{Q}) \approx 5.5 \cdot 10^3$);
10. an explicit Lipschitz constant of $q \mapsto W_q$ (ENC-1) for Step 5 with an estimated q;
11. ENC-T2(c′) in the principal class; VAR-P12(d); II-T2(b′) and Maxwell (Q6);
12. a ledger row for Trotter–Kato; the closure of the citations of the next section.

Citations to close. so that the steps "conditional on verification" become fully grounded: van der Put–Singer Prop. 1.34/Thm 1.27 (VAR-T6); DICT-B3; Whitney–Stein (DICT-ILIM, (H2)); Pazy ch. 5 (VAR-P11(e), VAR-L2-cor); Thaller ch. 5 (FW-1 with V); Lax–Oleinik = Kružkov [F1] (S4T-T3); Humphreys §4.1 Cor. B (P1-T4); Wei–Norman 1964, content (VAR-P8(b) "⇐"); Casas 2007 Thm 3.4 (VAR-T10(4)); Oberst 1993, full text; and the Phase 0–5 numbers not checked: Krantz–Parks Thm 2.3.5 (OP-T1), Cohen GTM 138 for (AC), Hörmander SCV 1.4.4, Runge in Rudin.

15.6 Statuses of the claims used by the algorithm

For completeness, the statuses recorded in the table "claims used → status" of the extended algorithm that are *not* fully proved: C3-A1 (written proof, not audited, no ledger row: every output that depends on it is "(A) conditional on C3-A1"); VAR-T6, DICT-B3, DICT-ILIM-Hs (candidate, later closed), VAR-L2-cor (not used), FW-1 with V, VAR-P8(b) "⇐", VAR-P11(e), VAR-T10(2)(4)(5), P1-T4 (Lie from a cited source), S4T-L2 and S4T-T3 (modulo [F1]), ENC-A9(b3) (cited source and example), and the five textbook examples (checked on examples only).

16 Directions for improvement and routes not taken

Starting from: the route registry `ROUTES.md` (decisions at Gates 1, 4a, 6 and 8), the research notes on the learning pipeline (`NOTE_ML`), the open questions of the extended algorithm, and the honest limits of the final report. *Why:* a record of what could make the work better: sharper constants, missing

audits, whole directions that were scouted and parked, and the statistical wrapper that was proposed and postponed. It also records which routes were eliminated and why, so that nobody repeats them. *Role:* companion to Section 15. That section lists open *questions*; this one lists *improvements and directions*.

16.1 Improving what is already there

Sharper encoder constants. The only proved bound on the Volterra encoder is $C_W = 1 + a^2 Q\, e^{a\sqrt{Q}}$; it overestimates by about four orders of magnitude at $a\sqrt{Q} = 5$ ($C_W^2 \approx 1.4 \cdot 10^7$ against $\cosh^2 \approx 5.5 \cdot 10^3$). A cosh bound or an exact Gronwall estimate would make the transported certificates of Step 5 useful in practice. An *explicit* Lipschitz constant of $q \mapsto W_q$ would allow Step 5 with an estimated potential.

Rigorous misfit. The misfit certificate is class (A) only with a rigorous inclusion of r_n (interval arithmetic or an error bound for the quadrature); with floating-point quadrature it is class (B). The numerically stable computation is the least-squares residual, not the Gram determinant ratio.

Cost model. No general cost estimate is proved for the dictionary levels other than the linear family ($(n+1)(n+2)/2$ monomials in $1+1$); the cost of rule R1 per level, and of the Goursat solve ($O(M^2)$ evaluations of q by iterated quadrature, indicative only), should be proved.

Audits not yet run. The elimination procedure C3-A1 has a written proof, no audit and no ledger row; every output that depends on it is "(A) conditional on C3-A1". Trotter–Kato convergence has no ledger row. The final text of this edition has had no proof-checker audit, no citation audit and no kill-argument review (the Phase 11 plan required all three).

Citations. Close the list of Section 15 (van der Put–Singer, Whitney–Stein, Pazy, Thaller, Humphreys, Wei–Norman, Casas, Oberst) and read the sources that unblock the "blocked" items; read Saut–Scheurer and Isakov to decide whether the unique-continuation lemma is already known; read Schindelar–Levandovskyy–Zerz in full to confirm the stated delta.

Rates under stronger hypotheses. Explicit jet thresholds beyond exponential-polynomials; explicit margins ρ_d for the heat and wave equations as functions of the domain; stochastic noise with rates in probability; non-integer smoothness in the Hölder estimates.

Dimension. The Volterra encoder is one-dimensional: in dimension $d \geq 2$ the characteristic problem is not of Goursat type. A genuine multidimensional analogue, or a proof that none exists with a local kernel, is open.

A first-principles proof of the Burgers closure. A written proof of the equivalence Oleinik $\Leftrightarrow$ Kružkov by vanishing viscosity on Cole–Hopf would remove the "modulo [F1]" from the shock results; the triage gives the plan.

16.2 Routes scouted and parked

During the project each candidate approach (a *route*) was scouted, then promoted, merged, parked or eliminated with a stated reason. The parked routes are the natural next directions.

Route	Content / why parked	Reopen when
RT-5 (symbol tensor)	the tensor keeps only the principal symbol (R6-P1); useful as a geometric invariant; reopened for systems (RT-SYS) and parked again	a use as an invariant "at infinity"
RT-R1b	needs a sign hypothesis on the datum	such a hypothesis is acceptable
Koopman / DMD	"dynamic mode decomposition is regression", excluded by the no-learning constraint; reopen with the normal form: eigenfunctions carry the nonlinear term	the no-ML rule is lifted
Painlevé, NLSPEC, HAANTJES	triggers in the verdicts of the nonlinear scouting; Painlevé kept as a *tool* (DICT-R4)	a nonlinear exact family needs it
RT-NA (non-autonomous propagators)	parked after Phase 8	triggers VAR-Q8, Q11, Q18
RT-ENC (the encoder as a route)	parked; lemmas moved to the dictionary and variable-coefficient routes	triggers ENC-Q4, ENC-Q10
RIGID, STOKES, CONV-BORDO	not touched in Phase 8	boundary identification with margin (CONV-Q16)
Weyl algebra for PDE	scouted only for ODE and A_2	a D-module factorisation tool is available
Inverse Sturm–Liouville (spectral data)	parked	spectral data become available as a datum
U3(a) kernels, U5	universality classes not covered	a use case
Homogenisation / gamblets	approximate, a tool (S) per scale, not an encoder	oscillating or rough V treated as noise
Wave operators $\Omega^*_\pm$	outside the bounded-domain model	scattering data are the datum
Hartogs $1+d$	P-convexity on non-convex domains in several variables (CONV-Q5)	non-convex $1+d$ decomposition needed
MON, NAT, MOB	minor routes, remain parked	—
Mori–Zwanzig, RT-R2a, POLE, gamblets as *methods*	eliminated: the criteria are false or only cited; the lemmas remain (Section 9)	—

Eliminated for good. Statistical consistency as a method (excluded by the no-learning rule), SPLIT-f, free probability, ICA, quiver representations, microlocal analysis beyond the characteristic variety, and Lie/Olver symmetry methods were eliminated or merged. RT-S1g was eliminated for the core (it gives no information for elliptic or parabolic operators) and kept for the hyperbolic part.

16.3 The statistical wrapper that was proposed and postponed

At the end of Phase 5 a learning pipeline was drafted around the exact analytic core (the note *Reading the structure of a differential equation from data*). It was *excluded* from the project

by decision of the user (no machine learning of any kind, including the autoencoder/mixture idea, "Idea I", postponed to a later phase). It is recorded here as a direction.

1. **Interpolate.** A learning algorithm fits a smooth function to noisy data (t_i, x_i, y_i) and returns not one function but a weighted family $(\hat{u}_s, \pi_s)$, with $\pi_s \geq 0$, $\sum \pi_s = 1$.
2. **Analytic core.** Each plausible function is passed to the exact algorithm, which returns its factors and the component each carries.
3. **Aggregate.** Weighted by the probabilities, this gives a distribution over factors (for example "transport 0.7 ± 0.1, diffusion 0.3 ± 0.1").
4. **Group.** The distribution sets the parameters of a standard clustering (how many groups, how spread out), cutting the data into groups that ideally contain one factor each, with a small "tangle number" (the Friedrichs cosine between factors; the size of E_{int} between terms).
5. **Repeat inside each group** until all groups are clean; the factors then rebuild the equation. The dashed loop of the pipeline: each group becomes a new data set.

The exact parts are the analytic core and the two tangle measures; the interpolation, the aggregation and the clustering were left as a proposal. The note asked two questions: whether the basis functions of the first step must be chosen in advance (and therefore only discover equations whose building blocks were anticipated), and how to avoid that choice. The answer built in Phase 8 is the filtered dictionary with a misfit certificate and a stopping rule (Section 11); the statistical aggregation itself remains open.

16.4 What was never done

- the catalogue of about 60 equation types (ODE, transport, diffusion, hyperbolic, integral and memory, beams and elasticity, quantum, non-linear) used as scouting material, and the tropical reading of scaling limits (only the dissipative-face estimate TROP-3′ and its corollary for advection–diffusion are proved);
- Lean sources and a Lean statement for the analytic parts (the Mathlib libraries are missing: no Weyl algebra, no C_0-semigroups);
- an independent human review: all reviewers were AI agents;
- the decision on global installation of the project skills and on Mathpix (Step 0 of the original plan, left to the user);
- the Phase 11 audits on the final text of this edition.

16.5 Honest limits of the whole project

The overall novelty is modest: many building blocks are classical (Malgrange–Ehrenpreis, the Nullstellensatz, Cole–Hopf, inverse scattering, Peetre, Volterra transformation operators, Oberst duality). The contribution is the package of characterisations, the negative theorems and the counterexamples, with verified statuses. The source policy (decision of the user, 26 October 2026) allowed results to rest on published theorems provided the source is verified: of 112 sources, 34 had the theorem number checked, 75 had the content checked (45 only in the metadata), none was not found, and three wrong citations were corrected (Humphreys Cor. B, Gel'fand–Dikii 1976, DLMF 9.7.10). Several load-bearing numbers were still unchecked at the end (Section 15). Work interrupted by API usage limits was resumed from the files with no loss recorded.

A Proofs

Throughout, $\Omega \subset \mathbb{R}^2$ is open and connected, $\mathcal{A}$ is $\mathcal{D}'(\Omega)$ or $C^\infty(\Omega)$, and $\mathcal{R} = \mathbb{C}[\tau, \xi]$ acts on $\mathcal{A}$ by $\tau \mapsto \partial_t$, $\xi \mapsto \partial_x$. For $z = (\tau_z, \xi_z) \in \mathbb{C}^2$, $\mathfrak{m}_z = (\tau - \tau_z, \xi - \xi_z)$ is its maximal ideal and $\psi_z(t, x) = e^{\tau_z t + \xi_z x}$.

A.1 Algebraic toolkit

Proof of Lemma 2.4. Coprime polynomials meet in finitely many points. Let $f, g \neq 0$ be coprime. By Gauss's lemma they are coprime in the principal ideal domain $\mathbb{C}(\xi)[\tau]$, so $af + bg = 1$ there; clearing denominators, $a'f + b'g = r(\xi) \neq 0$ with $a', b' \in \mathcal{R}$. At a common zero $r(\xi_0) = 0$, which leaves finitely many ξ_0; exchanging the variables, finitely many τ_0. Hence $V(f, g)$ is finite.

Splitting. Put $J = \{f/g : \ f \in I\}$ with $g = \gcd(I)$; it is an ideal and $I = gJ$, and no prime divides all of J. If $J \neq \mathcal{R}$, pick $0 \neq f \in J$ with distinct prime factors $\pi_1, \dots, \pi_r$ and, for each a, $g_a \in J$ with $\pi_a \nmid g_a$. The sets $\{c \in \mathbb{C}^r : \pi_a \mid \sum_b c_b g_b\}$ are proper linear subspaces, so some c avoids all of them; then f and $\sum_b c_b g_b \in J$ are coprime and $V(J) \subseteq V(f, \sum_b c_b g_b)$ is finite. By the Nullstellensatz a power of $\sqrt{J}$ lies in J, and $J = \mathcal{R}$ when $V(J) = \emptyset$. □

Proof of Lemma 2.7. ($\Rightarrow$) $Q(\partial)(q\,\psi_z) = (Q(z + \partial)q)\psi_z$ for every polynomial q, so the derivatives of $q\,\psi_z$ stay in the finite-dimensional space of polynomials of degree $\leq \deg q$ times ψ_z. Moreover $\prod_k \mathfrak{m}_{z_k}^{\deg q_k + 1} \subseteq \operatorname{Ann}(w)$, so $V(\operatorname{Ann} w) \subseteq \{z_k\}$; conversely $Q \in \operatorname{Ann}(w)$ forces $Q(z_k + \partial)q_k = 0$ by the linear independence of distinct exponentials, whose top-degree part is $Q(z_k)$ times that of q_k, so $Q(z_k) = 0$.

($\Leftarrow$) Let $\mathcal{R}w$ be finite-dimensional and p, q the minimal polynomials of ∂_t and ∂_x restricted to $\mathcal{R}w$, so that $p(\partial_t)w = 0$ and $q(\partial_x)w = 0$. On an open rectangle $I' \times J' \subset \Omega$, induction on $\deg p$ shows that every distribution with $p(\partial_t)v = 0$ is $\sum_{j,l} c_{jl}(x)\, t^l e^{\lambda_j t}$ with $c_{jl} \in \mathcal{D}'(J')$ and λ_j the roots of p (the base case uses that a distribution with $\partial_t V = 0$ on a rectangle is $c(x) \otimes 1$). Applying $q(\partial_x)$ and testing against functions dual to $t^l e^{\lambda_j t}$ gives $q(\partial_x)c_{jl} = 0$, so each c_{jl} is an exponential-polynomial of x. Thus w is an exponential-polynomial E near every point. The set where w coincides near y with a fixed E is open, and it is closed because two exponential-polynomials that agree on an open set agree everywhere; by connectedness it is Ω. □

A.2 Identification

Proof of Theorem 5.1. Let $I = \operatorname{Ann}(u) \ni P$ and write $I = G\,J$ as in Lemma 2.4. Since $P \in I$, $G \mid P$, so $G = c' \prod_i P_i^{f_i}$ with $0 \leq f_i \leq e_i$. First, $G \sim P$ iff $I = (P)$: if $G \sim P$ then $I = PJ \subseteq (P) \subseteq I$; if $I = (P)$, P divides all of I and lies in I.

(a), $\Leftarrow$ by contraposition. If $I \neq (P)$, some $f_i < e_i$, so $P/P_i = G\,H$ with $H \in \mathcal{R}$. A power $\mathfrak{n}^\ell$ of the ideal $\mathfrak{n}$ of the finite set $V(J)$ lies in J, and for $h \in \mathfrak{n}^\ell$, $h \cdot P/P_i = G(Hh) \in GJ = I$, i.e. $h(\partial)v_i = 0$. So $V(\operatorname{Ann} v_i) \subseteq V(J)$ is finite and $v_i \in \mathrm{EP}$ by Lemma 2.7.

(a), $\Rightarrow$ by contraposition. If $v_i \in \mathrm{EP}$, $V(\operatorname{Ann} v_i)$ is finite while $V(P_i)$ is infinite, so some $h \in \operatorname{Ann}(v_i)$ is not divisible by P_i. Then $Q = h \cdot P/P_i$ satisfies $Q(\partial)u = h(\partial)v_i = 0$; if $Q = PA$, cancelling P/P_i gives $h = P_i A$, a contradiction. So $Q \in I \setminus (P)$.

(b) $V(I) = V(G) \cup V(J)$ with $V(J)$ finite. If $f_i \geq 1$, $V(P_i) \subseteq V(I)$; if $f_i = 0$, $V(P_i) \cap V(I)$ lies in the finite set $\bigcup_{f_j \geq 1} V(P_i, P_j) \cup V(J)$, so $V(P_i) \not\subseteq V(I)$. Hence $V(I) = V(P)$ iff all $f_i \geq 1$, and $f_i = 0$ iff $v_i' \in \mathrm{EP}$ by the argument of (a) with $P/P_i^{e_i}$ in place of P/P_i.

(c) If $e_i = 1$ then $v_i = v_i'$. If P is monic of degree m in τ, divide any $Q \in \mathcal{R}$ by P in $\mathbb{C}[\xi][\tau]$: $Q = AP + \sum_{j<m} \varphi_j(\xi)\tau^j$ uniquely. Then $Q \in I$ iff $\sum_j \varphi_j(\partial_x)\partial_t^j u = 0$, and $Q \in (P)$ iff all $\varphi_j = 0$. □

Proof of Proposition 5.2. (a) If $I = (P)$, every annihilator is SP and $\deg SP \geq \deg P$ with equality iff S is a nonzero constant. (b) If $I = (P)$ then $G \sim P$ and $J = \mathcal{R}$, so $\Sigma_{\mathrm{p}}(u) = \emptyset$. Conversely, $V(J) = \emptyset$ gives $J = \mathcal{R}$ by the Nullstellensatz, hence $I = (G) = (P)$. (c) Let f_i be the multiplicity of P_i in G and $w = (P/P_i^k)(\partial)u$. If $k \leq e_i - f_i$, then $G \mid P/P_i^k$, say $P/P_i^k = GH$; for h in a power $\mathfrak{n}^\ell \subseteq J$ of the ideal of $V(J)$, $h \cdot P/P_i^k = G(Hh) \in I$, so $\mathrm{Ann}(w) \supseteq \mathfrak{n}^\ell$ and $w \in \mathrm{EP}$. If $k > e_i - f_i$ and $w \in \mathrm{EP}$, pick $h \in \mathrm{Ann}(w)$ with $P_i \nmid h$ (as in the proof of Theorem 5.1); then $hP/P_i^k \in I$, so $G \mid hP/P_i^k$, but P_i occurs in hP/P_i^k with multiplicity $e_i - k < f_i$, a contradiction. Hence $f_i = e_i - \max\{k : \ w \in \mathrm{EP}\}$. □

A.3 Decomposition and stability

Proof of Theorem 3.1. We prove the statement for open connected Ω with $H^1_{dR}(\Omega) = 0$, which holds for simply connected domains.

Step 1: finite modules have no obstruction. Let M be an $\mathcal{R}$-module of finite dimension over $\mathbb{C}$. Then $\mathrm{Ext}^1_{\mathcal{R}}(M, \mathcal{A}) = 0$ and $\dim \mathrm{Hom}_{\mathcal{R}}(M, \mathcal{A}) = \dim M$. By the long exact sequence it is enough to check one-dimensional modules $\mathcal{R}/\mathfrak{m}_z$, since M has a composition series with such quotients. A free resolution of $\mathcal{R}/\mathfrak{m}_z$ is the Koszul complex of $(\tau - \tau_z, \xi - \xi_z)$, so $\mathrm{Ext}^\bullet(\mathcal{R}/\mathfrak{m}_z, \mathcal{A})$ is the cohomology of $\mathcal{A} \to \mathcal{A}^2 \to \mathcal{A}$, $v \mapsto \big((\partial_t - \tau_z)v, (\partial_x - \xi_z)v\big)$, $(a, b) \mapsto (\partial_t - \tau_z)b - (\partial_x - \xi_z)a$. Multiplication by ψ_{-z} turns it into the de Rham complex of Ω with coefficients in $\mathcal{A}$, whose first cohomology is $H^1_{dR}(\Omega) = 0$ (for distributions by de Rham's theorem), and whose zeroth cohomology is the constants, of dimension one.

Step 2: two factors. For coprime Q_1, Q_2, $(Q_1) \cap (Q_2) = (Q_1 Q_2)$ by unique factorization, and $0 \to \mathcal{R}/(Q_1Q_2) \to \mathcal{R}/(Q_1) \oplus \mathcal{R}/(Q_2) \to \mathcal{R}/(Q_1, Q_2) \to 0$ is exact ($1 \mapsto (1, 1)$, $(a, b) \mapsto a - b$). Applying $\mathrm{Hom}_{\mathcal{R}}(\cdot, \mathcal{A})$, and using $\mathrm{Hom}(\mathcal{R}/(f), \mathcal{A}) = \ker_{\mathcal{A}} f$, gives the exact sequence $0 \to \mathrm{Hom}(\mathcal{R}/(Q_1, Q_2), \mathcal{A}) \to \ker Q_1 \oplus \ker Q_2 \xrightarrow{(u_1,u_2)\mapsto u_1+u_2} \ker Q_1 Q_2 \to \mathrm{Ext}^1(\mathcal{R}/(Q_1, Q_2), \mathcal{A})$. The quotient $\mathcal{R}/(Q_1, Q_2)$ is finite-dimensional by Lemma 2.4, so by Step 1 the last group vanishes: the sum map is onto, and its kernel has dimension $\dim \mathcal{R}/(Q_1, Q_2)$ and consists of solutions of a zero-dimensional ideal, i.e. of exponential-polynomials.

Step 3: induction. $P' = Q_2 \cdots Q_N$ is coprime to Q_1, so $\ker P = \ker Q_1 + \ker P'$, and by induction $\ker P' = \sum_{j \geq 2} \ker Q_j$. Dimensions of kernels add along the composition of the two sum maps, and $\dim \mathcal{R}/(Q_1, P') = \sum_{j \geq 2} \dim \mathcal{R}/(Q_1, Q_j)$ because $\mathcal{R}/(Q_1, Q_2Q_3 \cdots)$ is the direct sum of its localizations at the finitely many common zeros, where the intersection multiplicities add. This gives (7). □

Proof of Theorem 3.4. The sum operator $\mathcal{S}$ is bounded from $\bigoplus_j M_j$ into $\ker_{H^s} P$, and every u in its range has components in H^s. Its range is all of $\ker_{H^s} P$ iff the range is closed and dense. By the closed range theorem the range is closed iff $\mathcal{S}$ is bounded below on $(\ker \mathcal{S})^\perp$, i.e. iff $\gamma_N > 0$, and then the minimum-norm preimage satisfies (10). For two closed subspaces, $M_1 + M_2$ is closed iff $c_F < 1$, and $\gamma_2 = 1 - c_F$ [2]. In finite dimension every subspace sum is closed, and the sum equals the span of the sampled exponentials of P, each of which lies in the kernel of one factor. □

Proof of Proposition 4.2. For real $\zeta > c/\nu$ let $u_\zeta = e^{\lambda_\zeta t + \zeta x}$, $\lambda_\zeta = \nu\zeta^2 - c\zeta > 0$; then $Q(\partial)u_\zeta = 0$ and $\|u_\zeta\|_{H^s(\Omega)} \le C_s(1+\zeta)^{2s} e^{\lambda_\zeta T}\|e^{\zeta x}\|_{L^2(a,b)}$. With base time t_0 and horizon t, Theorem 4.1 gives

$$E_{\mathrm{int}} = e^{\lambda_\zeta t_0}\bigl(e^{-c\zeta t} - 1\bigr)\bigl(e^{\nu\zeta^2 t} - 1\bigr)\, e^{\zeta x},$$

whose $L^2(\Omega)$ norm is at least a constant times $e^{\nu\zeta^2 t}\|e^{\zeta x}\|_{L^2(a,b)}$ for large ζ. With $t_0 = 0$ and $t = T$ the ratio therefore grows at least like

$$e^{\nu\zeta^2 T - \lambda_\zeta T}(1+\zeta)^{-2s} = e^{c\zeta T}(1+\zeta)^{-2s},$$

which is unbounded for every s. For the second statement, $v + w$ is killed by Q and by $R = (\tau + c\xi)(\tau - \nu\xi^2)$. At a common zero of Q and R, either $\tau = -c\xi$, and then $Q = 0$ gives $\nu\xi^2 = 0$, or $\tau = \nu\xi^2$, and then $c\xi = 0$; in both cases $(\tau, \xi) = (0, 0)$. So (Q, R) has a single zero, and its common kernel is a finite-dimensional space of exponential-polynomials (Lemma 2.7). $\square$

B Glossary of classical tools

Each entry states a classical tool in one sentence and points to where this edition uses it.

Name	Statement in one sentence	Used in
Unique factorisation	Polynomials in two variables factor uniquely into irreducibles, up to constants.	Ch. 2
Bézout's theorem	Two curves without a common component meet in at most the product of their degrees points [7].	Ch. 2, 5.2
Hilbert's Nullstellensatz	A polynomial vanishing on the zero set of an ideal has a power in the ideal.	Ch. 5
Chinese remainder theorem	Coprime moduli split a quotient ring into a product.	Ch. 3
Malgrange–Ehrenpreis	Every nonzero constant-coefficient operator has a fundamental solution, and is onto on smooth functions on a convex set [4, 13].	Ch. 3, 5.2
Oberst duality	Solution spaces of constant-coefficient systems correspond to modules over the operator ring [16].	Ch. 5
Poincaré lemma	On a simply connected region every closed 1-form is exact.	Ch. 3
Friedrichs angle	$M + N$ is closed iff the cosine of the angle between M and N, after removing $M \cap N$, is smaller than 1.	Ch. 3.5
Miyadera–Voigt	Small relatively bounded perturbations of a generator generate a semigroup [6].	Ch. 4
Baker–Campbell–Hausdorff	$e^X e^Y = e^{X+Y+\frac{1}{2}[X,Y]+\cdots}$.	Ch. 4
Wei–Norman	Linear time-dependent systems on a solvable Lie algebra reduce to ODEs for scalar coefficients [18].	Ch. 4
Tychonoff	The heat equation has nonzero smooth solutions with zero initial datum [11].	Ch. 4
Cole–Hopf	$u = -2\nu\varphi_x/\varphi$ maps Burgers to the heat equation [8].	Ch. 6

Name	Statement in one sentence	Used in
Kružkov	Entropy solutions of scalar conservation laws are unique and L^1-stable [12].	Ch. 6
Inverse scattering	KdV and NLS are linearised by a nonlinear Fourier transform with trivial time evolution.	Ch. 6
Novikov	The stationary KdV hierarchy characterises finite-gap and soliton potentials [15].	Ch. 6

C Primer for readers new to the concepts

This appendix *adds* concrete explanations; nothing in the main text depends on reading it. It covers four notions in this order: ideals and annihilators (§B.1), exponential-polynomials (§B.2), the Chinese remainder theorem (§B.3), and the companion matrix on a numerical example (§B.4).

B.1 Ideals, annihilators and the gcd, starting from integers

Integers. The multiples of 6 form an *ideal* $(6) = \{\dots, -6, 0, 6, 12, \dots\}$: it is closed under addition and under multiplication by any integer. The ideal generated by 4 and 6 is $(4, 6) = (2)$, since $2 = 6 - 4$ and every combination is even: the ideal "remembers" only the gcd.

Operators. Replace integers by operators and "multiple of 6" by "kills u". For $u = e^x$ every operator $Q(\partial_x)$ with $Q(1) = 0$ kills u, i.e. every multiple of $\partial_x - 1$. So

$$\begin{aligned} \operatorname{Ann}(e^x) &= (\partial_x - 1), \\ \operatorname{Ann}(\sin x) &= (\partial_x^2 + 1), \\ \operatorname{Ann}(xe^x) &= ((\partial_x - 1)^2), \\ \operatorname{Ann}(e^x + e^{2x}) &= ((\partial_x - 1)(\partial_x - 2)). \end{aligned}$$

In one variable every annihilator ideal has a single generator (the *minimal annihilator*), exactly as every ideal of integers is (n). In two variables an ideal can need several generators, as $(\tau - \tau_0, \xi - \xi_0)$ does for $u = e^{\tau_0 t + \xi_0 x}$.

The gcd. $\operatorname{Ann}(u) = G_u J_u$ separates the part of the ideal that is a single polynomial multiple (G_u, "infinite part") from the part J_u that has only finitely many common zeros (J_u, "finite debris"). Example: for $u = f(x+t)$ with f generic (not an exponential-polynomial), $\operatorname{Ann}(u) = (\partial_t - \partial_x)$ by Theorem 5.1, so $G_u = \tau - \xi$ and $J_u = \mathcal{R}$; for $u = e^{x+t}$, $\operatorname{Ann}(u) = (\tau - 1, \xi - 1)$, so $G_u = 1$ and $J_u = \operatorname{Ann}(u)$.

B.2 Exponential-polynomials: examples and a test

Function	In EP?	Reason (derivatives of u span ...)
xe^x	yes	span$\{xe^x, e^x\}$, dimension 2
$x^2e^{3t} + \cos x$	yes	finite sum of polynomial $\times$ exponential ($\cos x = \frac{1}{2}(e^{ix} + e^{-ix})$)
e^{-x^2} (Gaussian)	no	$\frac{d^k}{dx^k} e^{-x^2} =$ (polynomial of degree k) e^{-x^2}: independent for all k, infinite dimension
$1/(1+x^2)$	no	derivatives are rational functions with growing denominators: infinite dimension

This is Lemma 2.7 in action: "$u \in$ EP" means "its derivatives span a finite-dimensional space".

B.3 The Chinese remainder theorem, concretely

Integers. The system $x \equiv 1 \pmod 2$, $x \equiv 2 \pmod 3$ has the unique solution $x \equiv 5 \pmod 6$: $\mathbb{Z}/6 \cong \mathbb{Z}/2 \times \mathbb{Z}/3$. Coprimality of 2 and 3 is what makes the pair of remainders determine x.

One-variable ODE. Let $p(\partial) = (\partial - 1)(\partial - 2)$. Then $\ker p = \ker(\partial - 1) \oplus \ker(\partial - 2) =$ span$\{e^x, e^{2x}\}$, and the Lagrange-type projectors are explicit: if $u = ae^x + be^{2x}$,

$$u_1 = \frac{\partial - 2}{1 - 2}\, u = \; a\, e^x,$$
$$u_2 = \frac{\partial - 1}{2 - 1}\, u = \; b\, e^{2x}.$$

This is Eq. (11) with ∂_t replaced by ∂.

Two variables. For the wave equation $Q_1 = \tau - \xi$, $Q_2 = \tau + \xi$ are coprime, but the ideal $(Q_1, Q_2) = (\tau, \xi)$ is not the unit ideal: the two lines meet at the origin. The quotient $\mathcal{R}/(\tau, \xi) = \mathbb{C}$ has dimension 1, which is the constant ambiguity $\delta(P) = 1$ of the wave equation. Theorem 3.1 says that this finite overlap is the *only* failure of the integer picture.

B.4 The companion matrix on a numerical example

For $u'' - 3u' + 2u = 0$ the companion matrix is $C = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}$, with characteristic polynomial $\lambda^2 - 3\lambda + 2$ and eigenvalues $1, 2$. The projectors of Eq. (4) are $\Pi_1 = \dfrac{C - 2\mathrm{Id}}{1 - 2}$ and $\Pi_2 = \dfrac{C - \mathrm{Id}}{2 - 1}$, and $e^{xC} = e^x \Pi_1 + e^{2x} \Pi_2$.

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